WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER #2 PROGRAM

AZNS – 127 (PAD #54), TYPE: IIF

Drilling Department

DECEMBER 2025

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TABLE OF CONTENTS

Contents

1. GENERAL INFORMATION 5
1.1 Field Introduction 5
1.2 Well Information 7
1.3 Area Distribution 8
1.4 Well Pad Specification 10
1.5 Reservoir Data 14
1.6 Formation Pressure Prediction in Block II 16
1.7 Drilling History 17
1.8 Workover #1 History 18
1.9 Casing & Liner Depth 37
1.10 Well Schematic before Workover 38
1.11 Well Schematic after Workover 39
2. GEOLOGICAL INFORMATION 40
2.1 Formation Stratigraphy and Lithology 40
3. LOGGING PROGRAM 41
4. PROGRAM OUTLINE 42
4.1 Contractors Involved 42
4.2 Time Breakdown 43
5. SEQUENCE OF OPERATION 45
5.1 Main Objective 45
5.2 Main plan 45
6. BHA PROGRAM 50
6.1 8-1/2" BHA 50
6.2 5-7/8" or 6" BHA 51
6.1 3 – 1/2" BHA 52
7. MUD PROGRAM 53
7.1 interval Description 53
7.1.1 Product Concentration & Function 54
7.1.2 Required Properties 55
7.2 interval Description 57
7.2.1 Solid – Free $CaBr_2$ System 57
8. COMPLETION 61
8.1 Proposed Completion string 61
9. BOP, WELLHEAD & X-MASS TREE 62
9.1 Existing Wellhead & X-Mass Tree 62
Wellhead Specification *63*
9.2 BOP 66
10. WASTE MANAGEMENT 67
11. HEALTH, SAFETY AND ENVIRONMENTS 68
11.1 WELL CONTROL 68

11.1.1 WORK Requirement of Well Control Equipment 68
11.1.2 Management of Well Control Equipment 68
11.2 Recording Reference 69
11.3 Flow Check Procedure 69
11.4 Primary Control Reference 69
11.5 Secondary Control Reference 70
11.6 Well control during drilling and completion operation 70
11.7 HEALTH 73
11.8 SAFETY 73
11.9 ENVIRONMENT 73
12. H_2S MONITORING & PREVENTION MEASURE 76
12.1 Monitoring H2S 76
12.2 Measures for H2S prevention 76
13. HANDOVER DATA 78
14. REQUIREMENTS 79
15. REVISION CORRECTIONS 81

1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and

Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-127 workover #2. Main headlines of program are de-complete the well and run new completion string.



Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Workover #2
Well Name / Type :	AZNS – 127 / Fahliyan well in Block II
Pad Number:	54
Well Type:	Side Track
Wellhead Coordinate:	N: 3,448,561 E: 196,167
TD Coordinate:	N: 3,448,538.7 E: 196,163
Well Objective:	De-complete the well, Re-complete the well
RTE:	12.95 m
GLE:	3.95 m
Well Duration :	45.9 (Main Plan) + 110.8 (Contingency Plan)
Total Depth (MD):	4140 m
Producer Formation:	Perforated in 4 – 1/2" liner 4120.7 – 4124.7 | 4117.7 – 4120.7 | 4088.2 – 4092 | 4075.7 – 4080.2 | 4071.2 – 4075.7 | 4060.7 – 4064.4 | 4052.45 – 4056.9 | 4048 – 4052.45 | 4023.4 – 4026.8 | 4000.55 – 4006 | 3995 – 4000.55 | 3989.65 – 3995.1 | 3983.3 – 3988.65
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1: AZNS-127 General Well Information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.

Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: AZNS-127 Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.

Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal

Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

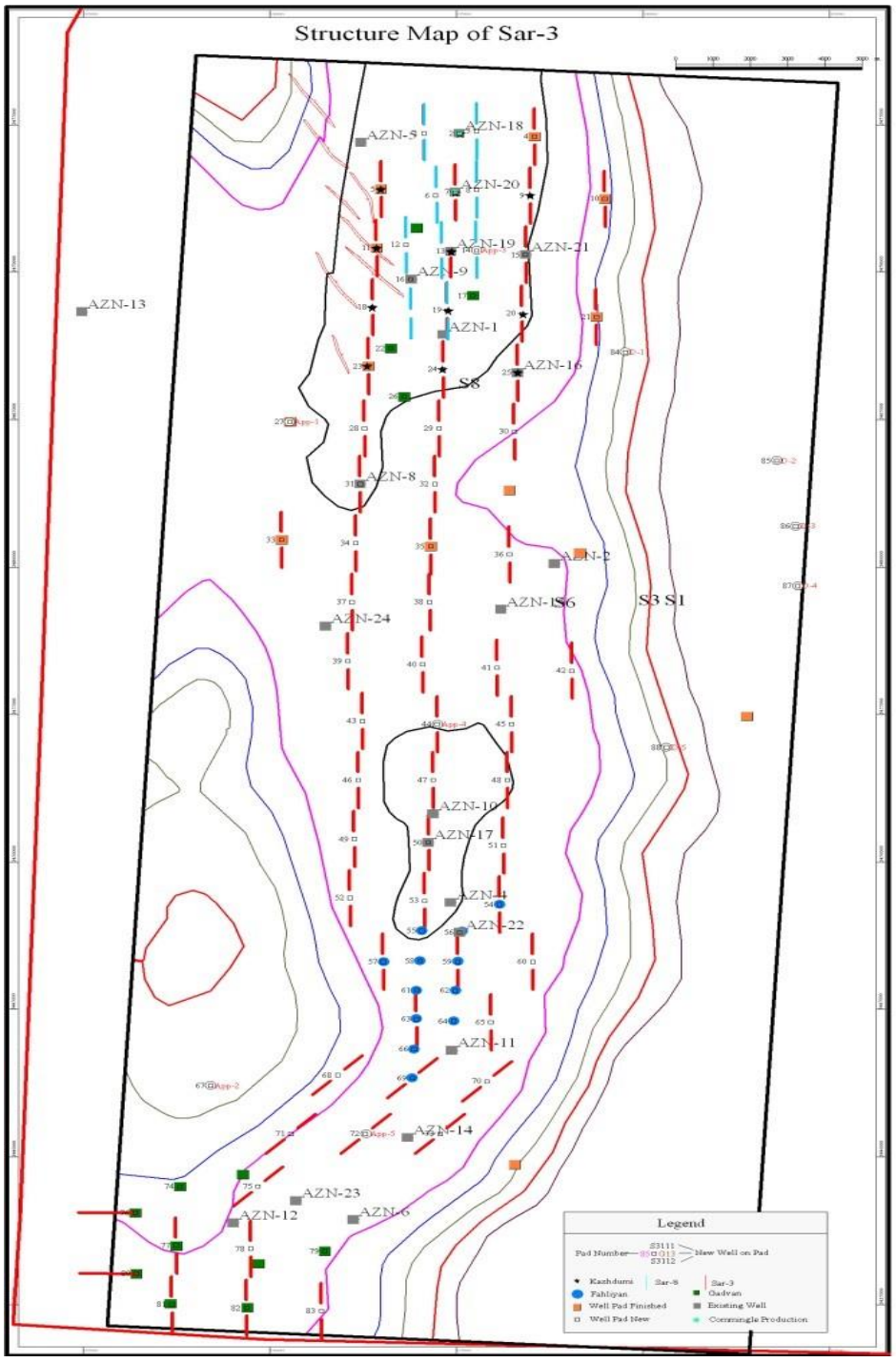


Figure 2: AZNS Area Distribution

1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

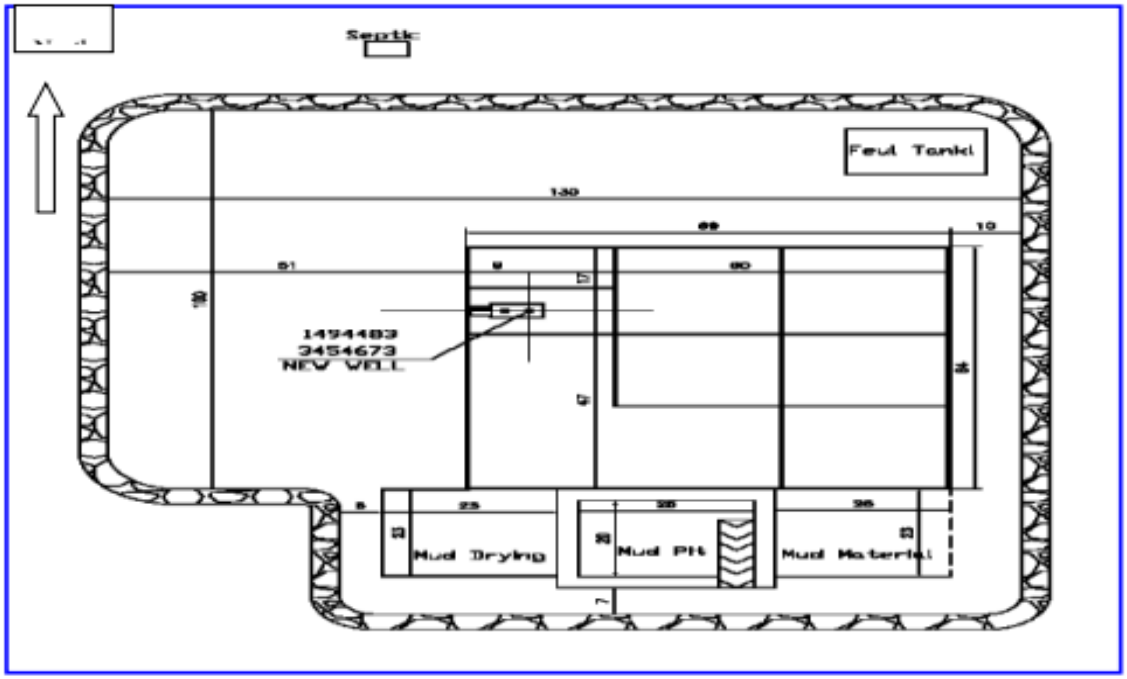
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2

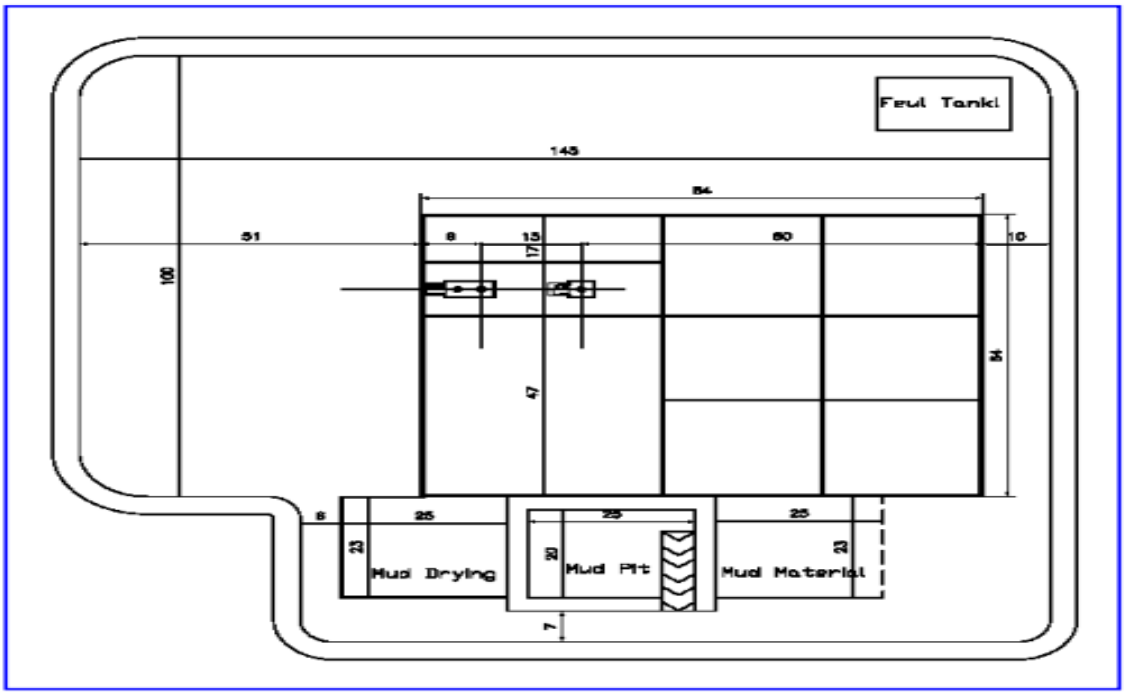
- Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
- Drainage: Width 30cm × Depth 30cm
- Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
- Cellar to cellar distance in multi-well well pad: 15m
- Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
- Mud material dimension is 25m*35m +granular surface
- Mud drying dimension is 25*35m + concrete surface
- Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

1. Types of well pads:

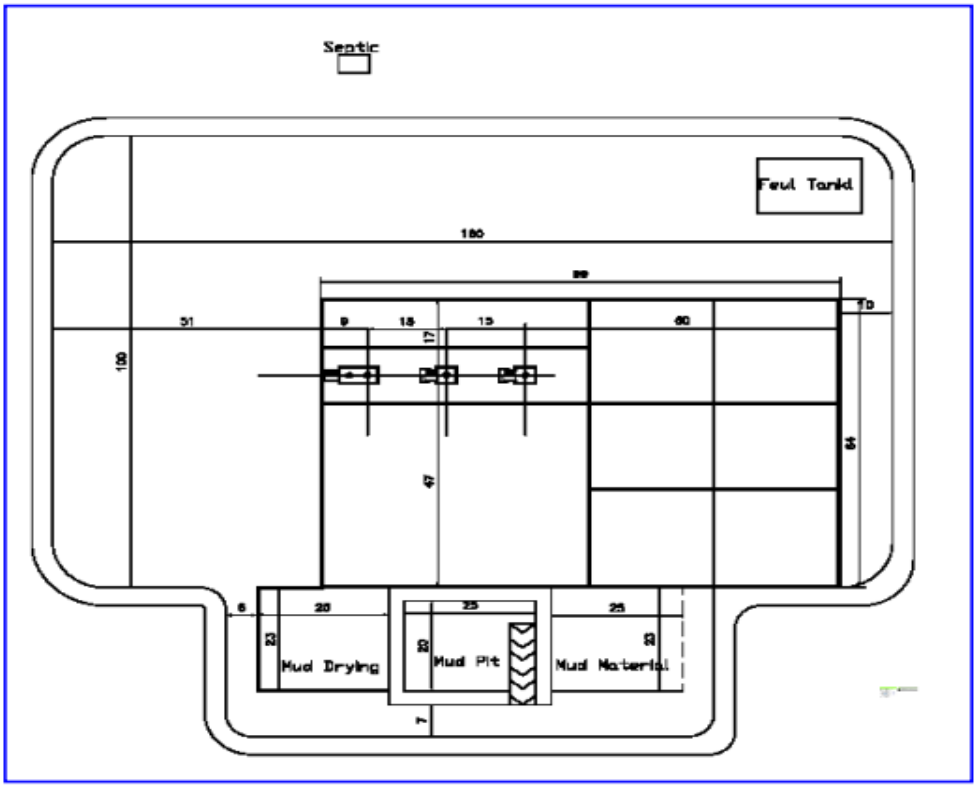
- One well in one pad.
- Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
- Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	-4000	M,TVDss
Datum Pressure:	Fah-4a= max 6800 psi Fah-5a= max 7400 psi	Psi
Pressure Gradient:	0.28	Psi/ft
Temperature:	132	C°
Temperature Gradient:	N/A	1.32 F/100ft 0.73 C/100ft 2.4 C/100m
H2S:	H2S in Gas at Atm condition 3000 PPM	PPM
CO2:	Co2 in Gas at Atm Condition 50000 PPM	PPM
Salt Salinity :	143000	ppm
Calcium Content:	8750	(mg/lit)

Table 4: Reservoir Data

Reservoir	Sub-layer	Mid of Sub-LayerDepth in AZNS-122 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
Fahliyan	Fah-1a	-3966.7	256.3	1.32	9100	MDT 195
	Fah-1b	-3983	257		9100	MDT 195
	Fah-1c	-4009	258		9200	MDT 195
	Fah-2a	-4035.7	259.3		8912	Dynamic Model
	Fah-2b	-4049.8	259.9		8924	Pressure Gradient
	Fah-3a	-4058	260.3		8440	Dynamic Model
	Fah-3b	-4067	260.6		8447	Pressure Gradient
	Fah-4a	-4077	261.1		7551	Dynamic Model
	Fah-4b	-4091	261.7		8100	MDT 195
	Fah-5a	-4108.6	262.4		8100	MDT 195
	Fah-5b	-4163.7	264.8		8100	MDT 195

Table 5: Predicted Pressure and Temperature and TTZ Contact

1.6 Formation Pressure Prediction in Block II

Predicted Formation Pressure in Block II				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-1095.9	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	1095.9-1461	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1461-1728	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1728-2196	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2196-2536	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2536-2626	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2626-2633	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2633-3278	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3278-3481	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3481-3647	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3647-3937	1.1-1.25	0.55-1.44	1.97-2.11
Fahliyan	3937-T.D.	1.60-1.3	1.20-1.75	2.08-2.12

Table 6: Formation Pressure Prediction

1.7 Drilling History

The AZNS – 127 well was drilled & completed by TDDC on date 1395.10.08 at 12:00 as describe below:

- 26" hole size drilled to 150^{m} w/ Mw 66 – 68^{pcf} & 20" casing ran to 146^{m} & cemented as program by 170^{bbl} cement.
- 17 – 1/2" hole size drilled to 1441^{m}, while increased Mw from 68 to 90^{pcf}, 13 – 3/8" casing ran & cemented to 1441^{m} by 780^{bb} cement.
- 12 – 1/4" hole size drilled to 3910^{m} w/ performance drlg & 3 – 15BPH loss. At depth 2354^{m} suddenly pressure dropped f/ 2850 to 1950^{psi} & string twist off from float sub & 9.18^{m} fish left in hole. Fish recovered by overshot 10 – 1/8". At depth 3117^{m}due to pressure dropped f/ 2560 to 2320^{psi} POOH string but string got stuck at 3040^{m} & free by applied 50^{klbs} overpull, POOH to surface & observed wash out on float sub above 8" PDM. Took MSFL/DLL/SONIC/CALIPER/GR/CDR & ZDL/CNC/DSL/WTS/TTRM logs, 9 – 5/8" casing ran & cemented to 3909^{m} with 1133^{bbl} cement.
- 8 – 1/2" hole size drilled to 4226^{m} with no major problem, took VDL/SBT/CCL/GR logs, string held up at 3905^{m}, tried to pass this point several times with no success, logged up from this pont to 100^{m} above 13 – 3/8" casing shoe. Due to tight hole Mw increased f/ 104 to 105^{pcf}. Took first full suite log as DLL/SONIC/MSFL/CDR from TD to 3810^{m} & second full suite log as CNT/LDT/NGS/GR from 3910^{m} to TD. 7" liner ran to 4226^{m}. Due to first dry test results, decided was made to tie back 7" liner to 3777^{m}. The well suspended with cement plug & completion string & rig released @1396.06.27.
- Re – entry the well at 1396.11.03 by retrieving completion string & ran new completion & rig released at 24:00 on 1396.12.23
- Rig less operation was performed on date 17.05.1398. tried to RIH with coiled tubing to unload the well. There was an obstruction in the well at depth 3918^{m}. Unable to pass the depth. Several runs were performed by coiled tubing and mill to remove the obstruction.
- Rig less operation for unloading and well flowing was performed on date 30.01.1400.
- Pressure monitoring was performed on tubing head spool vent port, it was 4,000^{psi}. Metal seal of tubing hanger is damaged.
- Try to run coiled tubing to TD and wash the wellbore with xylene. Run CCL/GR/MFC logging BHA in the well. It was observed 7" liner ID has been decreased to about 3inches in interval 3889 – 3911^{m}.
- Flow the well, an abnormal annulus pressure was observed.

1.8 Workover #1 History

The AZNS – 127 workover well was De-Complete by rig OEOC – 208 on Pad 54, from "08 – May – 2024" to "18 – March – 2024" as described in the following:

De – Completion Summary:

Installed 8" poor boy degasser flare. Pressure test SP & function test TDS, MPs; Ok. Checked section B: P-Seals & ring gasket; Ok. Checked section C: BTM P-Seal; Ok. Top P-Seal & ring gasket; Not Ok. Disconnected adjustable choke of production line. L/U pump truck to XMT kill line & annulus. L/U choke manifold to production line & annulus, meanwhile preparing 200bbl 106pcf mud. Pressure test all line after rectify several leakages. Open Ann & monitor Press; 4700 psi. Bleed off Press to 3000 psi, Build up to3800 psi in 1 Min. Increase Press f/ 3800 to 3900 psi in 5 Min.

P/U 60 Jts 5" DP Jt by Jt, made & R/B 20 Stds to derrick. Monitored annulus pressure build up to 4800psi after 6 hrs. Opened the annulus thru choke manifold to oil storage; after 6min. decreased pressure to 980psi. Decreased pressure gradually to 500psi in 20min. Closed the annulus; Shut in pressure 500psi. Built up to 1000psi in 15min. Opened the annulus, suddenly dropped to zero. After 6min closed the annulus. Opened the 2SV, top master valve & production valve. Tightened XMT cap. Connected hand pump to exit block, tried to open TR3SV. Monitored tubing pressure 3400psi. Opened the well f/ TBG to oil storage tank; dropped pressure to zero in 2min. Closed the well. Built up to 3700psi in 2min. L/U annulus to choke manifold. Obs. annulus pressure built up to 4300psi in 3 hrs. Opened the well & pressure decreased to 600psi. Closed the annulus. Pressure rapidly built up to 3300psi. Opened the well again. Pressure suddenly dropped to zero. Flew the well for 10min. Closed the well, built up to 3300psi. Monitored the well for1 hrs. Pressure built up to 4300psi.

Due to jammed stem of inside SOV of THS in close position, installed pressure gauge on annulus to choke manifold line & pressure test same to 7000psi, Ok. R/U slickline equipment. Installed slickline BOP & lubricator on XMT cap. Tried to pressure test, Obs. leakage between lubricator & BOP. Changed lubricator O-ring & tried to pressure test; No-Success. L/D previous BOP. Installed another slickline BOP & Lubricator on XMT cap. Pressure test slickline BOP & lubricator w/ 7000psi to 15min; Ok. RIH w/ 3.4" gauge cutter to TBG HGR, tried to pass this depth; No-Success. Pulled out to surface & R/D slickline BOP & lubricator.

Installed 4 – 1/16" 10k lubricator on XMT. Pressure test to 7000psi for 15min; Ok. RIH w/ BPV retrieving tools to TBG HGR & retrieved 4" BPV f/ TBG HGR. L/D 4 – 1/16" 10k lubricator. Due to unable to open TR3SV, decided M/U 5" DPs. R/D slickline equipment.

R/U slickline equipment. Pressure test; Ok. Fixed CL port exit block. Ran 3.4" gauge cutter to 206ft, RIH w/ 3.437" protection sleeve to 190ft, opened TR3SV while holding pressure to 1200psi. Applied 7500psi pressure in CL & opened TR3SV. Cont. running tools to 200ft, took weight, tried to set protection sleeve by jarring several times; No-Success. Flew the well with 1/4" choke size & Cont. work on set protection sleeve; No-Success. Pulled out tools to 191ft. Obs. stuck tools. Worked on stuck to max. 60lb over-pull; No-Success. Finally flew the well by annulus & tubing. Tools got free. Cont. pulled out tools to 50ft & stuck tools again. R/D lubricator. Obs. cut the slickline wire from 50ft & left in hole about 3m tools & 13m wire. RIH w/ 2.5" LIB to 157ft, POOH to surface; Obs. wire mark on same.

While wait on OEOC slickline service, RIH w/ 2.8" spear to TOF. Worked on fish & retrieved 2.45m of wire after 3 attempts. RIH w/ 2.8" LIB to 157ft, POOH to surface. Obs. rope socket marked on same. RIH w/ 2.8" blind box, POOH to surface. Rope socket marked on same. Welded 3.2" skirt on 2.2" JDC, RIH w/ JDC to TOF. Engaged w/ same. Tried to pull out, Obs. indication of shear JDC Shear pin. POOH to surface. Obs. sheared JDC Shear pin. Redressed w/ steel shear pin. RIH w/ JDC to TOF again. Engaged & worked on fish, Pulled to surface. Obs. retrieved 10ft stem of fish. RIH w/ 2.8" LIB to TOF, POOH to surface. Obs. slickline mechanical jar on the same. RIH w/ JDC to engage w/ jar. Worked on fish, jarring up. Pulling out fish 20ft. Cont. worked on stuck. RIH w/ blind box & knock on TOF, without progress. POOH to surface. RIH w/ JDC to engage w/ Jar. Worked on fish, jarring up. POOH to surface. R/D slickline equipment.

In order to work on fish w/ coiled tubing, wait on receive coiled tubing unit. While waiting on OEOC slickline service, R/U CT & M/U 2 – 1/8" dimple connector on 1 – 1/2" CT pipe & performed pull test to max 20klb. Obs. failure in dimple connector, disconnected same. M/U 1 – 11/16" roll on connector on 1 – 1/2" CT pipe & performed pull test to max 20klb, Ok. Installed CT riser on XMT & CT BOP. Close CT blind rams & tried to perform pressure test; No-Success. Tightened between two risers flange & tried to perform pressure test; No-Success. P/U riser & BOP; Tried to perform pressure test; No-Success. Changed 1 Jt CT riser & performed pressure test CT BOP & risers, Ok. RIH CT BHA to 48m & engaged w/ the fish; Obs. stuck the fish. Worked on stuck to max 18klb

over-pull; No-Success. While shearing JDC Shear pin @ 2 times & POOH to surface & redressed w/ brass shear pin.

In the 3rd attempt, caught the fish & worked on same upward to 18klb over-pull while bleeding off annulus pressure f/ 4800 to 2000psi. Obs. fish got free, jumped same & sheared the JDC Shear Pin. POOH to surface. Cut 12m 1 – 1/2" CT pipe & changed roll on connector. Performed pull test to 20klb; Ok. RIH w/ 1 – 1/2" CT & 3.2" JDC to 48m. Worked on fish & POOH to surface. In the last attempt caught, engaged the fish & bleed off annulus pressure f/ 4800 to 2000psi. Pulled out f/ 48 to 10m w/ 2 – 3klb drag. Obs. took over-pull. Worked on stuck upward to max 18klb; No-Success. Then Obs. CT got free & POOH to surface. Closed swab valve & unable close TMV. Obs. sheared JDC Shear pin & TOF located in TMV. Installed 1 No 4 – 1/16" 10k gate valve on XMT. Redressed JDC shear pin to steel & M/U on CT BHA. Tightened all bolts. Performed pressure test to 6000psi for 10min; Ok.

Due to refused work on fish & stopped operation from OEOC CT service, wait on CT Service to make a decision. Changed JDC shear pin to brass. RIH w/ 1 – 1/2" CT & 3.2" JDC to 11m & engaged w/ the fish. Worked on stuck upward to max 18klb over-pull; No-Success. Slack off 6klb weight on JDC & sheared JDC Shear Pin. POOH to surface. R/D CT injector head. R/U slickline equipment & worked on fish w/ JDC jarring up several times; No-Success.

Performed injectivity test & monitored the well & checked communication between string & annulus; Ok. Pumped & direct Circ. 30bbl fresh water by applying back pressure in order to check communication between string & annulus; Ok. Commenced killing the well by W&W method, "pumping pressure f/ 5500 to 3600psi, annulus pressure f/5500 to 2400psi". Meanwhile R/D slickline equipment & CT raiser.

In order to work on fish closed 3SV & released above pressure, M/U & RIH with 3 – 1/2" handmade JDC & tried to engage with fish several times, in the last attempt successfully engaged with fish & recovered w/ max 10klb over-pull. Operate 3SV; Ok. Monitored the well pressure, observed wellhead pressure decreased f/ 500 to 300psi. Bleed off to zero & remained constant. Annulus pressure decreased f/ 2800 to 2000psi, bleed off to 1800psi & remained constant. Performed cavity pressure test to 4000psi for 15min; Ok. Pumped & direct Circ. 125bbl 137pcf mud, "pumping pressure f/ 1300

– 2800psi to 1700psi & annulus pressure 2400psi". Got 78 to 85pcf gas cut mud in return. Stopped pumping.

Bleed off pressure f/ 2200psi to zero step by step & monitored the pressure for any build up in each step; Negative. Wellhead pressure decreased to zero & remained constant. Pumped & direct Circ. 125bbl 145pcf mud. Stopped pumping. Bleed off pressure f/ 700psi to zero step by step & remained constant. Pumped 20bbl 135pcf mud to annulus. Obs. wellhead & annulus pressure decreased to zero. Opened cap of 3/8" CIL & monitored for 1 hr. Ok. Cut the CIL. Closed 3SV & removed XMT cap. Installed TWCV in TBG HGR & test same to 5000psi for 15min; Ok. N/D XMT while passing CIL thru BTM section.

N/D XMT & N/U 13 – 5/8" 10k BOP stack. Connected hydraulic, choke & kill lines. L/U cement line to BOP & performed BOP pressure test as NIOC policy; Ok. Passed CL f/ BOP stack to rig floor. Installed bell nipple. Changed annulus to C/M line, tested same; Ok. P/U & M/U 5" DP, XOS (4 – 1/2" IF * 4 – 1/2" TPCQ), 4 – 1/2" TBG HGR running tools. Retrieved TWCV. Screw TBG HGR running tools + X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) + 1 Jt 5" DP + 1 No. Pup Jt on TBG HGR. Performed pressure test of running tool O-ring to 1000psi, Ok. F/U string, opened 3SV & monitored the well, Ok. Back off running tools & R/D same. Close 3SV. R/U Golpa flange, BOP & lubricator of slickline (OEOC Co.), performed pressure test same w/ 1000psi. Obs. leakage between Golpa flange & 4 – 1/16"gate valve. R/D 4 – 1/16" gate valve, changed ring gasket, pressure tested same again to 1000psi, Ok. Ran 3.4" gauge ring to 61m (3SV depth), Obs. drag @59m, pulled out to surface. Ran spear to 61m (3SV depth) to catch possible cable in the well, pulled out to surface, Obs. No cable. Ran w/ protection sleeve to 61m (3SVdepth), held 7500psi pressure inside CL, set protection sleeve inside 3SV, retrieved protection sleeve & pulled out to surface. B/O slickline BOP. R/U 4 – 1/16" gate valve on Golpa flange, M/U slickline BOP & lubricator & performed pressure test same w/1000psi, Ok.

Ran w/ protection sleeve to 61m, held 7500psi pressure inside CL, set protection sleeve inside 3SV & pulled out to surface. Ran w/ 2.230" gauge cutter to 61m (3SV depth), took weight, tried to pull out, Obs. slickline string stuck, tried to pass this point for several times, No Success. Pumped 5bbl 138pcf mud wile working on stuck, No success. Increased CL pressure to 10,000psi step by step, Obs. slickline string got free. Pulled out gauge cutter to surface & L/D same. Ran with GS pulling tools of protection sleeve & retrieved same. Pulled out protection sleeve to surface. Ran back in hole w/

protection sleeve to 61m (3SV depth) & tried to set protection sleeve for several times, No Success. POOH separation sleeve to surface. Decided to extend protection sleeve, RIH w/ new protection sleeve to 61m, set protection sleeve inside 3SV. RIH w/ 1 – 7/8" stem to 80m, checked opening of 3SV flapper, Ok, POOH stem to surface. RIH w/ 2.230" gauge ring to 3788m (first CIV depth), due to limitation of slickline cable tension decided to POOH to surface. B/O & L/D same.

R/D slickline BOP, lubricator, upper & lower sheaves. Installed X.O.S (4 – 1/16" flange * 4 – 1/2" IF box) on gate valve, pressure tested same w/ 1000psi, Ok. R/D mouse hole, R/U wireline equipment, P/U & M/U CCL & sinker bar. Pumped 100bbl 137pcf mud in string while applying 600 – 700psi pressure, back pressure in annulus. Obs. the well, Ok. B/O X.O.S (4 – 1/16" flange * 4 – 1/2" IF box) & cut box connection, welded 7" new Vam box connection on top of 4 – 1/16" flange & installed same on 4 – 1/16" gate valve. Installed wireline BOP & pressure tested same w/ 1000psi, Ok. Closed Hydrill, unscrew T.H.S tie down, P/U string, recorded string weight, applied allowable over pull to 298klb, got back on 211klb, according to limitation, applied 4klb.ft torque & tried to unlatch, No Success. RIH with CCL & sinker bar to 3850m (5m below PKR depth), POOH CCL & Sinker bar to surface & L/D same.

Held PJSM Assembled & armed colliding tools, RIH to 3850m, spaced out to 3845m, performed colliding, POOH colliding tools to surface, Obs. running tools of colliding tools & CCL left in hole. Pumped 115bbl 137pcf mud, Cont. pumping 685bbl 106pcf mud while dumping return mud, after pumping totally 800bbl mud, get 88pcf mud in return, divert return to poor boy degasser, Cont. pumping 106pcf mud, after pumped totally 850bbl mud, got 97pcf mud in return, all the time applied 700 – 1000psi back pressure in annulus. Opened T.H.S tie down screw, closed Hydrill, pressure tested tie down in open position w/ 1000psi, Ok. B/O & R/D 4 – 1/16" gate valve, Gol. Flang, landing Jt & TBG HGR running tools. P/U & M/U TBG HGR running tools, X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) & 2 Jts DP as landing Jt, screw running tools on TBGHGR, tried to release anchor seal with maximum 120klb overpull & applied 5klb.ft torque as forward for several times, No Success. B/O & L/D 2 Jts DP, X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) & TBG HGR running tools.

P/U & M/U TBG HGR running tools, landing Jt, Gol. Flange, 4 – 1/16" G.Flange & X.O.S (4 – 1/16" flange * 7" N. Vam), performed break Circ., NL-NG. Obs. the well, Static. R/U wireline equipment (Sealand Co.) & performed pressure test of wireline BOP w/ 1000psi, Ok. RIH w/ CCL & sinker bar to 3786m, took weight, tried to pass this point for several times, No Success. POOH to surface,

changed CCL, RIH w/ 1 – 7/8" CCL & sinker bar to 3786m, took weight, tried to pass this point for several times, No Success. POOH to surface. L/D wireline equipment. L/D 4 – 1/16" X.O.S (4 – 1/16" flange * 7" N. Vam), 4 – 1/16" gate valve, Gol. Flange, landing Jt & TBG HGR running tools, closed shear rams & performed break circulation, NL-NG. M/U TBG HGR running tools, X.O.S (4 – 1/12" TPCQ * 4 – 1/2" IF), 2 Jts 5" DP as landing Jt & screw on TBG HGR, at the beginning of P/U string, Obs. 270klb on weight indicator, while pulling the string, indication of EXP.Jt movement not observed (expansion joint is in full open position). Due to condition of EXP.Jt unable to follow releasing mechanism of Anchor seal due to TBG HGR is inside of THS unable to rotate.

P/U string 30cm to be sure TBG HGR is not inside THS (weight: 290klb), set torque limit on 7klb.ft, rotated & reciprocated string between 290 – 300klb weight on weight indicator to transfer TQ for several times, No Success. Released the TQ, P/U string to 400klb on weight indicator, to release Anchor seal, No success. Established circulation, Obs. gas cut mud in return, dumped 200bbl 50 – 55pcf cut mud, after 200bbl, Obs. 90pcf mud in return, divert return to poor boy degasser. Cont. circulation while getting 90 – 80pcf cut mud in return & applied 400 – 1400psi back pressure. Stopped operation due to poor performance of transfer pump of middle, degasser, desander, desilter tank & also hopper #1 & #2 causing unable to transfer mud to suction tank. Worked on same to rectify problem while holding 1200psi back pressure. Resumed Circ. while applying 1200 – 1400psi back pressure, got 106pcf mud in return. Stopped pumping, Obs. the well, Static. Worked on collided point to get string free with Max. 120klb overpull & applying 6klb.ft TQ as forward, string got free, string weight on weight indicator: 195klb. R/D wireline, R/U CSG running equipment, POOH completion string while L/D landing Jt, TBG HGR, fluted swage, 3SV w/ long & short flow coupling (Obs. protection sleeve was not set in profile) & totally 35 Jts 4 – 1/2" TBG, meanwhile Circ. across wellhead, NL-NG.

Cont. POOH completion string while L/D 94 Jts 4 – 1/2" TBG (totally 294 Jts), while using safety clamps & Circ. across wellhead; NL-NG. Obs. last Jt of string w/ underneath coupling washout, worn & torn in coupling area. Also Obs. 5 No. of cross coupling protectors left in hole & 9 No. of cross coupling protectors parted because of eroded condition. After Jt No. 35 some of cross couplings had been lost, eroded & parted. Obs. some of asphaltene on cross coupling protectors of injection line & slips area of 4 – 1/2" TBGs so need to be washed with steam, which lead delay in POOH completion string.

P/U & M/U 7 – 7/8" Over shot (w/ 4 – 3/4" Basket Grapple Mill Control, 4 – 1/2" Basket Grapple) & Fishing BHA & RIH to 1880m, NL-NG. Estimated TOF: 2706m. Took weight @ 1880m, tried to pass several times; No Success. Established Circ. & pass gently. Cont. wash down to 2660m while trying to pass several intervals w/o wash down; No Success (Asphaltene precipitation indication); NL-NG. Meanwhile got 150bbl 50pcf gas cut mud & dumped same @ 2020m. Established Circ. to uniform Mw in/out on 106pcf, got 300bbl 75 – 95pcf gas cut mud w/ NL-NG. Obs. huge amount of asphaltene precipitation on shakers. Resumed wash down to 2731m while took weight several depth & pass same. Confirmed top of fish @2731m. Tried to catch the fish several times; No Success. Commenced POOH w/ 7 – 7/8" over shot & fishing BHA f/2731m to surface. Circ. CWH; NL-NG. Obs. some scratches on outer body & edge of over shot guide.

R/U slickline equipment & ran 3.8" magnet to 2731m, held up. POOH to surface. 1lb junk recovered (mixed with asphaltene). R/D slickline equipment while waiting on receive 8 – 3/8" Mill. P/U & M/U 12 Jts 6 – 1/2" DC, 3 Jts 5" HWDP & RIH OEDPs to 1250m Jt by Jt. Circ. CWH; NL-NG. POOH string to surface & L/D same. RIH OEDPs by 12 Jts 4 – 3/4" DC & 12 Jts 3 – 1/2" DP. POOH OEDPs to surface & R/B. While Circ. CWH; NL-NG. Difficulty B/O & L/D 1 Jt 5" Pup Jt, 5" FOSV & Circ. swage due to over torque. Tried to B/O 11" plug tester from 1 Jt 5" DP w/ 48klb.ft TQ several times; No Success. Set 11" wear bushing in THS.

P/U & M/U 8 – 3/8" M.M J.Mill, 1 No. 6 – 5/8" Junk Sub, 1 No. 7" Junk Sub, 6 – 1/2" D.Pin, 6 – 1/2" Bit Sub & RIH w/ 4 Stds to 2729.5m & took weight on this depth. Recorded off BTM parameters. Commenced milling obstruction f/ 2729.5 to 2736m; NL-NG. While got 100bbl 80 – 90pcf gas cut mud @ BTM's up. Cont. milling f/ 2736 to 2739m; NL-NG. Circ. BTM's up, over Circ. till clean shakers w/ Hi-Rate; NL-NG. Obs. metal wool on ditch magnet & also CIL cover on shakers. Worked on junk sub several times. Commenced POOH w/ 8 – 3/8" M.M J.Mill & milling BHA f/ 2739 (TOF) to surface. Circ. CWH; NL-NG. Mill Condition: 80% used. Obs. mark of TBG as top of fish on mill. Both junk subs full of junk including pieces of CIL, cross coupling & metal wool. Recovered 19lbs junk with 2 junk subs.

P/U & M/U 7 – 7/8" overshot, dressed with 4 – 1/2" Basket Grapple & 4 – 3/4" mill control & RIH to 2739m. Worked on Fish & tried to catch & engage with fish several times. Obs. 20 – 50klbs excess weight on M/D & suddenly drop excess weight. Commence POOH w/ 7 – 7/8" overshot & fishing BHA f/ 2739m (TOF) to surface, Circ. CWH; NL-NG. Obs. mark of TOF on 3 wickers of basket

grapple & also mark of junk on outer body of guide. P/U & M/U 7 – 5/8" overshot (dressed w/ 4 – 1/2" Basket Grapple, 4 – 1/2" Mill Control & Extension sub) & RIH to 2739m. While W/D last 2 Stds, tried to catch & engage with fish. P/U string Obs. caught & engaged w/ fish. Worked on fish w/ 280 to 450klbs on M/D several times; No Success. Commence POOH overshot to surface. Obs. fishing jar mandrel packing was failed. R/U tubing running equipment, Obs. top of CILs were collapsed, cut same & bleed of pressure. Released overshot & L/D 6-1/2" DC 7-7/8" overshot. Obs. washed out TOF, 4-1/2" basket grapple, 4-1/2" mill control, inside 7-7/8" bowl & broken 1 No. basket grapple wicker.

POOH completion string totally L/D 37 jts 4-1/2" tubing. Obs. twist off string f/ tubing coupling & recovered all cross couplings & CILs. TOF depth at 3078m. Performed static flow check. P/U & M/U 8-3/8" M.M.J.mill used & RIH to 1170m, performed static flow check. Closed annulus preventer & Cir. Thru C/M & uniform Mw in/out on 106pcf, got 30bbl 50 – 90pcf gas cut mud while monitored pit level w/ NL.NG. Opened the well & performed static flow check, Ok. Resumed RIH 8-3/8" mill & milling BHA to 3078m. Commence milling f/ 3078.2 to 3079.3m while Obs. metal wool on magnet. Performed static flow check, Ok. Commenced POOH milling BHA to surface.

P/U & M/U 7-5/8" overshot & RIH to 3079.3m. Established Cir. & CBU w/ hi – rate, huge amount of asphaltene observed on shakers & got 20bbl 102pcf gas cut mud. Tried to catch & engaged with fish. P/U string & Obs. the fish was caught. Worked on fish w/ maximum 390klbs on M/D. H/L dropped to 245klbs. Commenced POOH fishing BHA to surface w/ NL.NG. Changed TDS elevator & POOH completion string & L/D 8 jts 4-1/2" tubing. Obs. twist of string f/ tubing coupling & also Obs. tubing couplings was worm & eroded. TOF at 3152.5m.

P/U & M/U 7-7/8" overshot & RIH to TOF at 3152.5m. Caught the fish after several attempts, worked on fish w/ maximum 480klbs on M/D. Applied different parameters several times, Obs. H/L dropped to 245klbs. Commenced POOH string & fishing BHA to surface. Obs. 1 joint 4-1/2" tubing retrieved & twist of tubing coupling. Released overshot f/ 4-1/2" tubing.

P/U & M/U 8-3/8" M.M.J.mill used & RIH to 3159m. Took weight at this point, tried to wash down several times, no progress. Wash & ream down to 3161.2m. commenced milling from 3161.2 to 3162.6m while got 100bbl 90 – 100pcf gas cut mud w/ NL.NG. Obs. metal wool on ditch magnet &

Asphaltene on shakers. Recovered 20lbs metal from magnet. Worked on junk sub several times & commenced POOH milling BHA to surface. 5lbs metal, debris & Asphaltene f/ 2 junk subs.

P/U & M/U 7-5/8" overshot & RIH to 3162m. washed down last stand & CBU w/ NL.NG. tried to catch the fish. Obs. catch & engage w/ fish. Worked on fish w/ max 400klbs on M/D, assembled 2-1/8" CCL, 2-1/8" weight bar & 1-11/16" Wt. bar. R/U wireline equipment, installed top sheave on monkey board & installed BTM sheave to mouse hole. M/U -1/8" CCL, 2-1/8" weight bar & 1-11/16" Wt. bar on rope socket & passed same thru string. RIH CCL & Wt. Bar as dummy run to 3170m. took weight, tried to pass this point several times, No success. POOH to surface & R/G equipment. R/U CT & M/U 1 roll on connector & XOS & Golpa flange. M/U XOS on string & CT BOP & performed pressure test. RIH CT to 3610m while Obs. drag at 3158m & 3213m, cleared same. Took weight at 3610m, tried to pass this point several times w/ applying max. 2000lb weight, No success. POOH CT to surface.

R/U Sealand wireline equipment. RIH wireline 2-1/8" CCL & 2 No. 2-1/8" weight bar as dummy run to 3222m, Obs. drop tension. Passed this point & RIH to 3226m, Obs. drop tension. Tried to pass this point several times, No success. POOH to surface & L/D assembly. Ran back wireline w/ 1-11/16" CCL & 3No. 1-11/16" weight bar as dummy run to 3612m, took weight & tried to pass several times, No success. POOH & L/D assemblies. Assembled & RIH wireline w/ 1-3/8" FPIT & CCL tools to 3600m meanwhile checked the string w/ applied 170klbs overpull. POOH wireline assey. to surface & L/D same. FPIT results as bellow:

- 1000 & 2000 & 3100m: 100% free
- 3200 & 330m: 50% free
- 3400m: 25% free
- 3420m: 10% free
- 3440 & 3450m: 5% free
- 3460, 3470, 3490, 3500 & 3600m: 0% free

Assembled & RIH w/ 1-7/8" colliding & CCL tools to 3612m. Obs. drop tension. POOH to 3594m & correlated depth w/ CCL, applied 180klbs overpull on string. Shot the colliding, checked & worked

on string, No success. POOH wireline to surface. Again assembled & RIH w/ 1-7/8" colliding & CCL tools as run #2 to 3398m while correlating depth. Applied 180klbs overpull on string & shot the colliding. No indication Obs. worked on string, No success. POOH wireline to 3275m, wireline got stuck, worked on stuck several times to get free. Continued POOH to surface & L/D colliding Assy. Worked on string upward w/ max. 180klbs at neutral point several times, No success. Assembled & RIH wireline w/ 2-1/8" CCL & 1 No. 2-1/8" weight bar as dummy run to 3272m, Obs. drop tension, tried to pass this point, No success. POOH to surface.

RIH w / 2-1/8" colliding & CCL tools as run #3 to 3290m, Obs. drop tension, POOH to 3140m & ran back to 3281m, correlated depth & applied 200klbs overpull on string, shot colliding. Obs. indication of shot, POOH wireline to surface & L/D wireline equipment. Worked on string upward w/ max. 160klbs overpull f/ neutral point to max. 50klbs overpull in several times, No success. RIH wireline w/ 2-1/8" colliding & CCL tools as run #4 to 3185m, correlated depth & applied 200klbs overpull on string, shot the colliding & Obs. indicator. POOH to surface & L/D equipment & R/D unit. Worked on string w/ max. 160klbs overpull upward & downward several times, string got free. POOH string f/ 10m while Obs. 80klbs overpull & abnormal high pressure. CBU & balanced Mw in/out, Obs. huge amount of debris & tubular coating & Asphaltene on shakers, got 300bbl cut mud min 92pcf. POOH to surface, Obs. 7-5/8" overshot & retrieved 21.1m of 4-1/2" tubing f/ collided point & L/D same. Estimated TOF at 3183.1m.

P/U & M/U new 8-3/8" M.M.J. mill & RIH milling BHA to 3182.5m. Tagged TOF. Commenced milling the fish f/ 3182.5 to 3185.8m. worked on junk sub several times, pumped 60bbl hi vis pill & CBU till cleaned up shakers. POOH milling BHA to surface. P/U & M/U 7-7/8" overshot & RIH to tag the fish at 3185.8m, CBU (pick – up weight: 246klbs, Slack weight: 242klbs). Tried to engage with fish, Obs. overpull. Worked on string by applying different parameters several times, string got free. Commenced POOH the string & fishing assembly to surface. Obs. 7-7/8" overshot with 4-1/2" tubing fish neck + 8 Jts 4-1/2" tubing + 73.318 + 9.268m of 4-1/2" tubing w/ CIV control line protector & L/D same. (TOF at 3270.77m)

P/U & M/U 8-3/8" M.M.J. mill & RIH to 3203m, took weight, tried to pass this point several times, No success. W & R/D to 3269.5m while CBU, Obs. 65pcf cut mud w/ NL.NG. Tagged TOF & commenced milling the fish f/ 3269.5 to 3274m. worked on junk sub several times, pumped hi-vis pill & CBU w/ NL.NG. POOH Milling BHA to surface & L/D same.

P/U & M/U 7-7/8" overshot & RIH to TOF at 3274m. Engaged with fish & applied different parameters till string got free. POOH overshot to surface. Obs. 4-1/2" tubing fish neck, 6.7m + 2 Jts 4-1/2" tubing (18.58m) + cut coupling 4-1/2" tubing (0.14m) w/ CIV control line protector & L/D same. P/U & M/U 8-3/8" used M.M.J mill & RIH to 3297.5m. Took weight, tried to pass this point several times, NO success. Tag the fish at 3297.5m & commenced milling the fish to 3298.8m. POOH milling BHA to surface & L/D same. Retrieved 6lb junk while Circ. cross wellhead. P/U & M/U 7-5/8" overshot & RIH to TOF. Caught the fish & worked on string by applying different parameters several times.

R/U Sealand wireline unit & RIH w/ 2-1/8" CCL & 1 No. 2-1/8" weight bar & 1 No. 1-11/16" weight bar as dummy run to 3318, 3333 & 3385m. Obs. drop tension at same depths. Several tries to pass depth 3386m without any succeed, POOH wireline to surface & L/D Assy. Ran wireline 2-1/8" colliding & 1-11/16" CCL to 3385m, Obs. drop tension, correlated the depth & applied 180klbs overpull on string. Fired the colliding shots. POOH to surface & L/d equipment. Worked on string to get free. POOH 60m of string w/ 7-5/8" overshot to surface & L/D same.

P/U & M/U used 8-3/8" M.M.J mill & RIH to 3260m. Washed down to 3300m, hard wash down to 3306m while string stall several times. Continued medium wash & ream down to 3384m & hard wash & ream down to 3390m while got 240bbl gas cut mud & huge amount of Asphaltene & CIL cover on shakers. Commenced milling obstructions f/ 3390 to 3391m w/ NL.NG. POOH the milling BHA to surface & L/D.

P/U & M/U 7-5/8" overshot & RIH to 3391m to TOF, tried to engage with fish several times, No success. Commenced POOH the string & fishing BHA to surface. P/U & M/U used 8-3/8" M.M.J. mill& RIH to BTM. Took weight at 3390.7m, commenced milling f/ 3390.7 to 3391.9m, POOH 8-3/8" milling BHA to surface & L/D same. P/U & M/U 7-5/8" overshot & RIH to tag the fish at 3391.9m. Due to tubing & casing annulus filled with junk & Asphaltene, unable to guide overshot into the fish. Commenced POOH overshot & fishing BHA to surface. P/U & M/U used 8-3/8" M.M.J. mill & RIH to 3391.9m. Commenced milling the fish to 3392.7m, worked on junk several times & POOH milling BHA to surface. P/U & M/U 8-3/8" M.M.J. mill & 9-5/8" scraper & RIH to 3350m. POOH milling BHA & scraper to surface. R/U wireline equipment & assembled 3-5/8" CCL-GR-CAST. RIH logging equipment to 3350m & logged up to 35m. POOH to surface while logging f/

3300 to 3150m, R/D wireline equipment & RIH w/ OEDP to 3390m while L/U CMT line & prepare for CMT job.

Side Track 8 – 1/2" Hole Size Summary:

L/U CMT line to string. Pumped 8bbl 75pcf brine as spacer ahead. Mixed & pumped 105sxs (SV: 25.4bbl, SW: 118pcf) CMT "G" Delijan, pumped 1bbl 75pcf brine as spacer behind, displaced w/ 170bbl 108pcf mud. WOC to get hard, performed static flow check. POOH w/ OEDPs. P/U & M/U 8-1/2" TCI bit & RIH to 3130m. Took weight, W & R/D f/ 3130 to 3170m & W/D to 3310m, CBU, Obs. CMT slurry on shakers. Continued W/D to 3347m, took weight. D/O hard CMT to 3350m. Performed positive pressure test by applying 1000psi for 15min. Ok. Commenced POOH 8-1/2" TCI bit & D/O BHA to surface. (bit dull grade: 1-1-WT-A-E-X-NO-BHA).

P/U & M/U used 8-1/2" TCI bit & 9-5/8" scraper & RIH to TOC at 3350m. scrap f/ 3130 to 3350m, 3 times each stand. Pumped & Circ. out 50bbl hi-vis pill till shakers clean. POOH 8-1/2" TCI bit & 9-5/8" scraper to surface & L/D. (bit dull grade: 1-1-WT-A-E-X-No-BHA)

P/U 1 joint 5" DP & M/& 9-5/8" hydraulic whipstock Assy. & ran to 20m, M/U UBHO & align to whipface. Continued run whipstock Assy. to 3340m while F/U string w/ fresh mud & passed drift thru all tubulars. R/U wireline logging equipment & RIH w/ gyro tools 3317m (UBHO depth). Oriented whipface on 90 deg. POOH gyro tools to surface & R/D wireline equipment. L/U CMT line to string & performed pressure test to 3500psi, Ok. Set whipstock packer at 3339.08m (bottom of whipstock Assy. at 3340m) w/ 3500psi stepwise. Bleed off pressure & performed 75klbs overpull to shear the screws on the rugby mill, L/D CMT line. (top of whipface: 3333m, Btm of Whipface: 3337m). Applied 2000psi pressure through the string for breaking 2 No. of nozzles plug, Ok.

Commenced milling as side track #1 with controlled parameter f/ 3333.3 to 3337.2m w/ NL.NG. Continued milling to 3341.3m. Confirmed KOP at 3338.6m by geologist. B/R & R/D window interval f/ 3333.3 to 3341.3m several times to smooth. Pumped 30bbl hi-vis pill & CBU till shakers clean. Commenced POOH w/ 8-1/2" TCI bit & L/D same.

P/U new DHM Lang Fang, adjusted AKO 1.5 deg. P/U & M/U used 8-1/2" PDC bit, 6-1/8" UBHO & 6-1/2" NMDC, performed shallow test of tools, Ok. RIH to 1500m, while fill up string every 10 STDs, w/ NL-NG. Performed shallow test (w/ GPM: 385, SPP: 1170psi) & took signal, Ok. Cont.

RIH to 3337.4m (BTM of Whipface @ 3337.6m), Took weight, POOH to 3332m (Top of Whipface @ 3333.3m), Turned string 90 Deg to right, ran back to 3339.5m, took weight, POOH to 3338.5m, pumped 20bbl Hi-Vis Pill, RIH w/ pumping (GPM: 450, SPP: 2030psi) & rotation (RPM: 20) to 3339.5m, took weight, W&R to 3341.30m, w/ NL-NG. Commenced directional drlg as rotation mode f/ 3341.3 to 3383.5m. Due to Obs. drop inclination, decision was made to POOH, stopped drilling, POOH to 18m, while Circ. cross wellhead, w/ NL-NG. Performed pressure test of 6-3/4" D.H.M & MWD tools w/ 8-1/2" PDC Bit (GPM: 400, SPP: 800psi), Obs. No Signal of MWD. L/D MWD tools, POOH to surface. L/D 8-1/2"PDC Bit, dull bit grading: (1-2-BT-C-X-I-WT/HC-BHA).

P/U & M/U used 8-1/2" B.Bit #02 RR3 & RIH w/ last 6-3/4" D.H.M, P/U new 6-3/4" MWD tools & RIH to 18m, align tools, P/U 6-3/4" Gyro Sub, align tools, performed shallow test (GPM: 400, SPP: 800psi & GPM: 500, SPP: 1200psi), Ok, Cont. RIH directional BHA w/ gyro sub to 3379m. took weight, W/D to BTM at 3383.5m while took survey at 3354.3m (Inc.: 1.1 deg, Az.: 161.4 deg) & 3369.7m (Inc.: 0.9 deg, Azi.: 145.9 deg). POOH to 3377.5m, about 6m off bottom. R/U logging equipment, RIH gyro tools as wireline to 3358m (gyro sub depth), adjust tools face of DHM. POOH wireline to surface & R/D logging equipment. RIH to BTM, resumed drlg as slide mode f/ 3383.5 to 3390.7m. Continued drlg as slide mode & rotation to 3415.5m. CBU till shakers clean. Commenced POOH on elevator to 3330m, with good hole condition. Cont. POOH to surface. L/D jar & gyro sub & directional BHA.

P/U new 6-3/4” DHM, adjusted AKO 1.5 deg. P/U & M/U used 8-1/2” PDC bit, M/U 6-3/4” float sub + 6-1/2” UBHO +6-1/2” NMDC, performed shallow test of BHA, Ok. Performed drlg assembly offset. RIH to 3415.5m while W/D last stand & filled up every 10 stds. Commenced drlg as rotation & slide mode f/ 3415.5 to 3454m while W & R/d every last drilled std before connections. Continued drlg as rotation & slide mode to 3522m. POOH with B/R to as wiper trip to 3400m with good hole condition. W & R/D to BTM. In order to drop inclination commenced drlg as slide mode f/ 3522 to 3524.5m w/ NL.NG. Continued drlg as rotation as slide mode to 3666.5m. While drlg string got stall & stuck several times in sliding mode drlg. POOH with B/R as wiper trip to 3530m & RIH back to BTM with good hole condition. Pumped & spotted 70bbl inhibited hi-vis pill & POOH on elevator to surface while Circ. cross wellhead w/ NL.NG & L/D BHA.

M/U used 8-1/2” PDC bit & drlg BHA w/ 8-1/2” stabilizer & RIH to 3445m. Took weight at this depth. RIH with R/D to 3448m, string got stuck, worked on stuck by jarring up w/ max. 150klbs

overpull to get free. Continued RIH with hard R/D to 3666.5m w/ NL.NG. commenced drlg to 3781m w/ NL.NG. String got stuck while B/R last drilled stand. Worked on stuck by applying torque & jerk down to get it free. Stopped drlg due to fault in TDS (frequent rotation loss while drlg). RIH back to BTM & resumed drlg f/ 3781 to 3835m w/ NL.NG. Commenced POOH drlg BHA to surface while CBU cross wellhead w/ NL.NG. L/D same. (bit dull grading: 2-5-BT-N-X-1/32”-CT-PR)

P/U & M/U new 8-1/2” PDC bit & RIH with drlg BHA to 3750m with good hole condition. Washed & R/D to BTM at 3835m & clear tight spots. Commenced drlg with low parameters to 3891m. CBU in order to take last sample while B/R last drilled stand w/ NL.NG. TD confirmed at 3891m. Soft B/R as wiper trip f/ TD to 3840m, soft R/D to BTM. Circulated out 40bbl hi-vis pill to clean shakers. Spotted 130bbl inhibited hi-vis pill w/ NL.NG. Commenced POOH 8-1/2” drlg BHA on elevator to surface with good hole condition & L/D same. (bit dull grading: 1-1-BT-N-X-I-NO-TD)

7” Liner Operation Summary:

Closed SBR & VBR top rams, retrieved wear bushing, set plug tester & performed BOP test as NIOC policy, Ok. R/U 7” liner running equipment. P/U & M/U 7” liner equipment & RIH to 49m while M/U swage & break Circ. to confirm float function for flow passing thru, Ok. Continued RIH to 746m w/ NL.NG. Install 7” * 8-1/2” spiralizer. P/U & M/U rotational hydraulic 7” liner hanger Assy., while installing wiper plug on slick joint f/ LG, Ok. M/U circulating swage & established Circ., check POB sealing, Ok. Filled up tie back sleeve w/ oil & grease. M/U TDS on setting tools neck & continued RIH to 3890.5m. Recorded P/U & S/O string weight on M/D (320 / 290klbs). Dropped ball, chased the ball to seat in LDC & increased pressure to 1800psi. held pressure & set 7” liner hanger at 3890.5m by applying 70klbs weight & applied totally 100klbs weight.

L/U CMT line, M/U CMT head on string. L/U CMT line to CMT head, P/U string, increased pressure to 3100psi, Obs. shear ball seat & got returned. Released setting tools, P/U string 0.8m performed pull test & confirmed string is free, M/U CMT head on string. L/U CMT line to CMT head, P/U the string & established circulate thru CMT head manifold by rig pump up to 900psi, while preparing CMT solution. Performed pressure test on CMT line & pump truck & batch mixer, Ok. CBU w/ NL.NG. Pumped 10bbl spacer ahead by rig pump, pumped 10bbl chemical wash, mixed & prepared 397sxs (SV: 95bbl, SW: 118pcf) CMT “G” delijan, pumped 95bbl CMT slurry 118pcf. Displaced CMT with totally 260bbl mud. Obs. shearing after pumped 177bbl mud & 2600psi. Obs. bumping at

3000psi w/ NL.NG. Held pressure as integrity test for 10 min. Ok. Released pressure & check back flow, No return. B/O CMT line & P/U string for 1m, slack off 30klbs on tie back sleeve& set liner hanger top packer. L/D CMT head & check string weight, Ok.

POOH 12 stds to 2788m, flushed the string. Continued POOH 5 more stds. CBU w/ NL.NG. Performed static flow check, resumed POOH w/ 7" liner hanger setting tools to surface. Filled up the well, L/D liner hanger setting tools. Set 11" wear bushing in T.H.S, P/U & M/U 8-1/2" TCI bit Kingdream & RIH to 3127m, took weight at this depth, washed down w/ 320 GPM to 3129m, tried to wash down w/ 10klbs weight without any succeed. Commenced D/O soft to medium CMT f/ 3129 to 3139m, Cont. D/O medium to hard CMT to 3149.5m, confirmed 7" liner lap at this depth. Pumped 50bbl Hi-Vis Pill & CBU till the hole got clean w/ NL-NG. POOH 8-1/2" TCI bit & D/O BHA to surface & L/D same w/ NL-NG. Bit dull grade 1-1-WT-A-E-I-No-BHA.

P/U & M/U & RIH 6-1/8" MT bit new Kingdream & C/O BHA to 3150.5m w/ NL-NG. while washed down, pass 7" liner lap & D/O hard CMT & 7" POB f/ 3150 to 3152.5m, dressed & smoothed POB interval without rotation several times. Continued RIH to 3784m, while washed & reamed down to 3840m, D/O POB, D/O hard CMT to LDC depth at 3853.5m. Pumped 30bbl Hi-Vis Pill & CBU till the hole got clean. Commenced POOH 6-1/8" MT bit & C/O BHA to surface & L/D w/ NL-NG. Bit dull grade 7-2-BT-(N, M)-E-I-ER-BHA.

P/U & M/U 6-1/8" used Kingdream w/ scraper & RIH to BTM at 3853.5m while scraped inside 9-5/8" CSG & 7" liner several times. POOH to surface, B/O & L/D bit & scraping BHA.

R/U & assembled Sealand Co. CCL/GR/SBT/VDL logging tools. RIH wireline logging tools to 3145m & logged down to 3852m. Logged up f/ 3845 to 3050m. Ran back to 3450m to repeat the log. Logged up f/ 3450 to 3050m. Powered off tools & POOH to surface, R/D Sealand logging tools & equipment.

Dry Test Summary:

P/U & M/U OEOC 9-5/8" dry test tools for 7" liner lap & RIH to 30m above liner lap at 3120m. Used 62.4pcf water as cushion to 2627m. R/U dry test surface equipment & L/U to CMT line. Recorded P/U & S/O string weight 230/225klbs, rotated string 6 rounds forward, S/O 25klbs & set 9-5/8" dry test packer at 3119m. Obs. hydro-spring tester valve opening indication on M/D. Monitored

open period 3hrs. Obs. no bubble & annulus level was static. S/O weight & rotated string 11 rounds forward closed DCIP valve. Recorded close period for 6hrs. & Obs. no bubble. Established reverse circulate & got water cushion as same as F/U volume. Continued reverse circulation till Mw uniformed in & out to 88pcf & secured the well. After test Obs. pressure increase to 213 psi. Unset 9-5/8" dry test packer w/ normal P/U weight. B/O & L/D dry test surface equipment, meanwhile M/U 5" circulation swage & established circulate w/ NL-NG. Performed static flow check for half an hour, Static. Commenced POOH 9-5/8" dry test assy. to surface, B/O & L/D tools.

6 – 1/8" Hole Size Summary:

Changed BOP top VBR, pressure tested same to 7500psi for 10min. Ok. Set 11" wear bushing in T.H.S. P/U & M/U 6" M.M.J. mill & RIH to 3853m. Commenced D/O LDC & hard CMT f/ 3853 to 3891.5m. Commenced drlg formation w/ 6" M.M.J. mill 3893.5m. Smoothed 7" liner shoe several times. Performed flow check, Static. POOH milling BHA to surface & L/D same w/ NL-NG.

P/U & M/U 6-1/8" new PDC bit Beste & RIH to 3871m, connected TDS & established circulation & displaced the well with fresh 100pcf KCl-Poly-CaCO3 mud until Mw uniformed on 100pcf w/ NL-NG. L/U CMT unit to circulation swage, closed top rams & performed FIT by applying pressure thru string to 7500psi step by step. Monitored & Recorded pressure on choke manifold. FIT @3871m, EMW: 108pcf

Resumed RIH to BTM, pumped Hi-Vis Pill & commenced drlg formation f/ 3893.5 to 4003m while B/R & R/D last drilled stand. Obs. gas cut mud & max 82% total gas. Stopped drlg & decision was made to increase Mw to 104pcf by added kill mud to system. Resumed drlg formation f/ 4003 to 4035m w/ NL-NG. POOH w/ hard B/R to 4014m. string got stuck at 4022m & made it free by applying different parameters. Continued POOH w/ hard B/R to 3870m w/ NL-NG. RIH back while R/D & B/R every stand & checked intervals without RPM to TD. Resumed drlg formation f/ 4035 to 4140m as TD.

Pumped 30bbl Hi-Vis Pill & circulated same out. POOH as condition trip with B/R to 3891m (7" liner shoe) & performed static flow check, Static. RIH back on elevator to BTM with good hole condition. Pumped & circulated out 30bbl Hi-Vis Pill. Pumped & spotted inhibited Hi-Vis & Hi-Dens fresh mud at BTM. POOH to surface & L/D 6-1/8" bit & BHA, bit dull grade: 0-0-NO-A-X-I-NO-TD.

P/U & M/U 4-1/2" liner assy. included float shoe, liner joints, float collar, LDC & RIH to 62m. F/U string & continued RIH to 353m. P/U & M/U PPA rotational hydraulic set/Mechanical release liner hanger assy. included tie back sleeve, LG & top packer, POB & rotational hydraulic hanger. Recorded P/U & S/O hook load 12/12klbs, F/U tie back sleeve w/ oil & grease & Cont RIH 4-1/2" liner on 3-1/2" & 5"DP to 4140m (btm) Set hanger. M/U CMT Head on string. L/U CMT Line to CMT Head, L/U CMT Line to CMT Head. Perform cmt job as per plan. POOH to surf. M/U 6" MMJ Mill RIH w/ 6" MMJ Mill and RIH to 3480m, W/D to 3720m. D/O cmt to 3780.5m. Circ. Btm's up. POOH to 200m while L/D excess tubulars till, Cont. POOH f/ 200m to Surf. L/D excess tubulars in derrick. RIH w/ 3-1/2" J.Mill on 3-1/8" DC & 2-7/8" DP Jt by Jt to 254m, Cont. RIH w/ 3-1/2" J.Mill on 3-1/8" to 3780.5m (4-1/2" L.Lap). commence D/O med. Cmt f/ 3780.5 to 3784m, 'D/O POB, Cont. wash and ream & Clean out inside 4-1/2" LNR to 4030m, Cont. wash and ream & Clean out inside 4-1/2" LNR to 4085m. D/O Med to Hard CMT f/ 4085 to 4102m. Tag LDC @ 4102m. Pump & Circ. Out Hi-Vis Pill. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 4102 to 1150m, 'Cont. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 1150m to surface, RIH w/ open end string & 7" Scraper BHA to 917m, Due to change in plan from client side, POOH w/ OEDPs to surface, RIH w/ Clean out BHA to 450m, Cont. RIH w/ Clean out BHA to 4100m, Cont. Wash & Ream Dn to 4102m, D/O Pump Dn Plug, Wiper Plug & 4-1/2" LNR LDC. 'Cont. D/O Hard CMT to 4134m, Circ. Out Hi-Vis Pill, Spot 20 bbl Non-Solid Brine, POOH to 3100m, Cont. POOH w/ 3-1/2" M.M J.Mill & Clean out BHA f/ 3100m to surface, Change top rams to 4-1/2" x 7" VBR, Press test same, Run CCL/GR/RBT Wireline logging tools to 4120m, POOH to 1900m, Cont. POOH w/ CCL/GR/RBT Wireline logging tools to surface. Run CCL/ 2-7/8" Gun as dummy run to 4120m. RIH w/ 3-1/2" Mill & Clean Out BHA to 2550m, String plug, work on same & Unplug, RIH w/ clean out BHA to 4134m, Circ. Out tandem pill, Spot Non-Solid Brine, POOH to 3350m, Cont. POOH w/ 3-1/2" M.M. J.Mill & Clean out BHA f/ 3350m to surface. Run CCL & 2-7/8" Blank Gun as dummy run to 4134m. RIH w/ 2-7/8" WCP HSD Gun Assy as Run #1 to 4124.7m. Fire the gun. POOH to 100m. Wait on daylight. Perform 2nd run of perforation f/ 4120.7 - 4117.7m to & f/ 4092 to 4088.2 m, Wait on daylight, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 4 to 3587m, Try to pass this depth, RIH w/ Clean out BHA to 1600m, Meanwhile Slip & Cut Drlg line @ 450m, Cont. RIH w/ Clean out BHA f/ 1600 to 4128m, CBU, Spot Non-Solid Brine, POOH to 1750m, Cont. POOH w/ 3-1/2" M.M J.Mill f/ 1750m to surface, Run dummy run to 4080.2m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #4 to 4080.2m. Fire the gun. POOH to 100m. Wait on daylight, Cont. wait on daylight. Resume POOH

W/L w/ WCP HSD gun assembly f/100m to surface, run dummy run to 4075m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #5 & #6 as per plan. Fire the gun. POOH to Run #6 to 100m. Wait on daylight, 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run #7 & #8 as per plan. Fire the gun. POOH to Run #8 to 100m. Wait on daylight. 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 9 & # 10 as per plan. Fire the gun. POOH to Run # 10 to 100m. Wait on daylight. 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, run dummy run to 4060m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 11 & # 12 as per plan. Fire the gun. POOH to Run # 12 to 100m. Wait on daylight. 'Cont wait on daylight. Resume POOH W/L w/ WCP HSD gun assembly f/100m to surface, run dummy run to 3989.65m, RIH w/ 2-7/8" WCP HSD Gun Assy as Run # 13. Fire the gun. POOH to surface. R/D WL unit. Press. Test BOP. P/U and M/U 3-1/2" J.Mill and RIH to 4050m Cont. RIH w/ 3-1/2" M.M.J.Mill and RIH to 4134m (btm) Circ. Btm's up. Spud 25 bbl Comp. fluid @ btm. POOH to 2300m, Cont. POOH f/2300m to surface while L/D exceess tubulars. Perform CCL logging. RIH & L/D 6 Jts 3-1/2" DP & 15 Jts 4-3/4" DC, 'RIH w/ 7" Scrapper to 3286m. Scrap PKR interval 3286 to 3325m. Cont RIH to 3777m. Pump Hi-Vis. Circ. Btm's up, Pump Hi-Vis. Circ. Btm's up. Pump 60 bbl Hi-Vis pill (104pcf, Ferobar free) + 20 bbl Hi-Vis pill (82 pcf brine) followed by Synthetic Completion Fluid and displace the well, Obs. The well, static. POOH and L/D DPs jt by jt to 2700m, 'Cont. POOH w/ 7" scrapper while L/D Jt by Jt f/ 2700 to 770m, Cont. POOH w/ 7" f/ 770m to surface, R/U slick line & TBG running equip. Start RIH 3-1/2" X 4-1/2" comp string to 73m, Cont. RIH 3-1/2" X 4-1/2" comp string to 406m. Run 2.8" gauge ring. Work on rectify lakages of pump truck lines, Cont. RIH 3-1/2" X 4-1/2" comp string to 1470m. While press. Test string @ 406 & 1224m; OK, Cont. RIH w/ Totally 284 Jts 4-1/2" TBG (18.9#, L-80, HSM-3) & 22 Jts 3-1/2" TBG (10.2#, L-80, HSM-3) to 2932m. While Press test completion string @ 2010 & 2932m; "OK". 'Run completion string to 3257m, M/U TR3SV Assy & Perform P/T TR3SV, RIH to 3274m, Perform P/T Completion string & inflow test of flapper, Cont. Run to 3325m, Perform P/T completion string, 'Change L/J, P/T Completion string; OK. Set Protection Sleeve. Ret. check valve f/ SSD. Set Plug in No-Go. Due to Client Request, decided to repeat string Press test. Ret. Plug f/ No-Go. Set Check valve in SSD. P/T Completion string; OK. Ret. check valve f/ SSD. RIH w/ Plug to top of SSD. In order to record upward Wt.; Obs. Suddenly drop tension. POOH to surface; Obs. About 830m Slick line wire, rope socket, 12 ft stem, 6 ft Mech. Jar, running

tools & 2.75" Plug left in hole. Due to lack of stem; wait on Rec. stem. 'Run LIB, Run 2.8" Wire grab w/o no catch wire, Ret. P.Sleeve, Run 3.2" Wire grab; Recover 2 ft wire, Run 3.2" 3 leg wire grab; Recover fish w/o plug, Set P.Sleeve, Run 2.8" Wire grab. 'Ret. Plug, Run Gauge ring to 3330m, Set Plug & Prong in No-Go, Run Shifting tool to SSD, Set PKR, Perform Pull test, Land TBG HGR, P/T String to 5000 psi, 'P/T tie downs, Ann, Ret. P.Sleeve, Back off & L/D L.J, Install TWCV & P/T same against SBR, N/D BOP Stack, N/U BMV & Related P/T, 'N/U XMT, P/T same, Ret. TWCV, Set P.Sleeve, Ret. Prong & Plug, Ret. P.Sleeve, Install BPV, Close All XMT Valve except BMV & Secure the well. Released the rig.

1.9 Casing & Liner Depth

Casing Size	Hole Size	Depth		PPF	Grade	Collapse (Psi)	Burst (Psi)	Tensile strength (Klb)
		From	To					
20"	26”	Surface	146	94	K – 55 BTC	520	2110	1480
13 – 3/8"	17-1/2”	Surface	1441	68	N – 80 BTC	2260	5020	1556
9 – 5/8"	12-1/4”	Surface	2732.64	53.5	T – 95 TPCQ	7340	9680	1477
9 – 5/8"		2732.64	3909	53.5	T – 95 TSH Blue	7340	9680	1477
7" Liner	8-1/2”	3147	3891	29	110 SS TPCQ	8530	11220	929
4 – 1/2 Liner	6-1/8”	3780.4	4139.5	13.5	110 S TPG2	10680	12410	422

Table 7: AZNS-127 Casing & Liner Data

1.10 Well Schematic before Workover

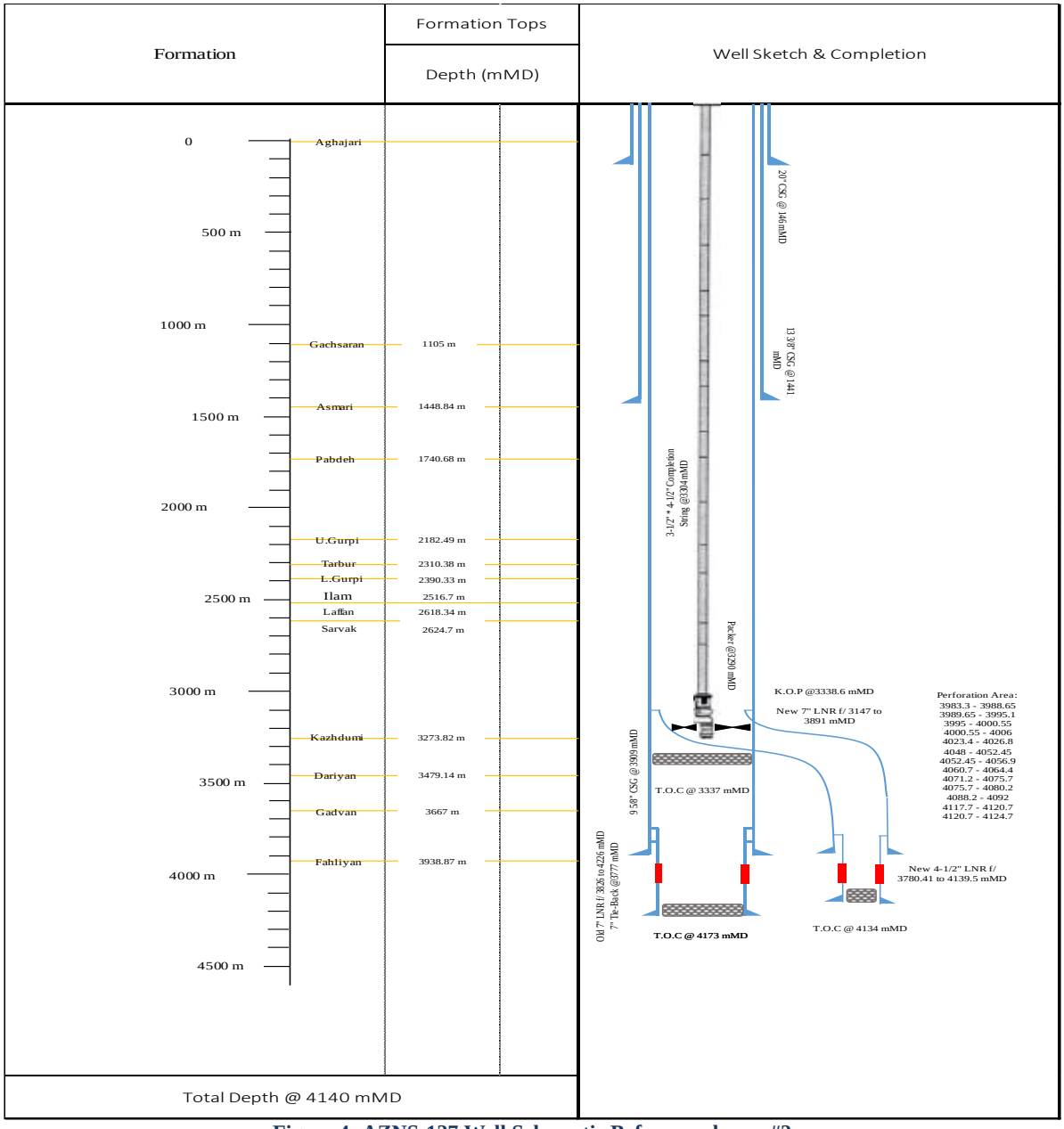


Figure 4: AZNS-127 Well Schematic Before workover #2

1.11 Well Schematic after Workover

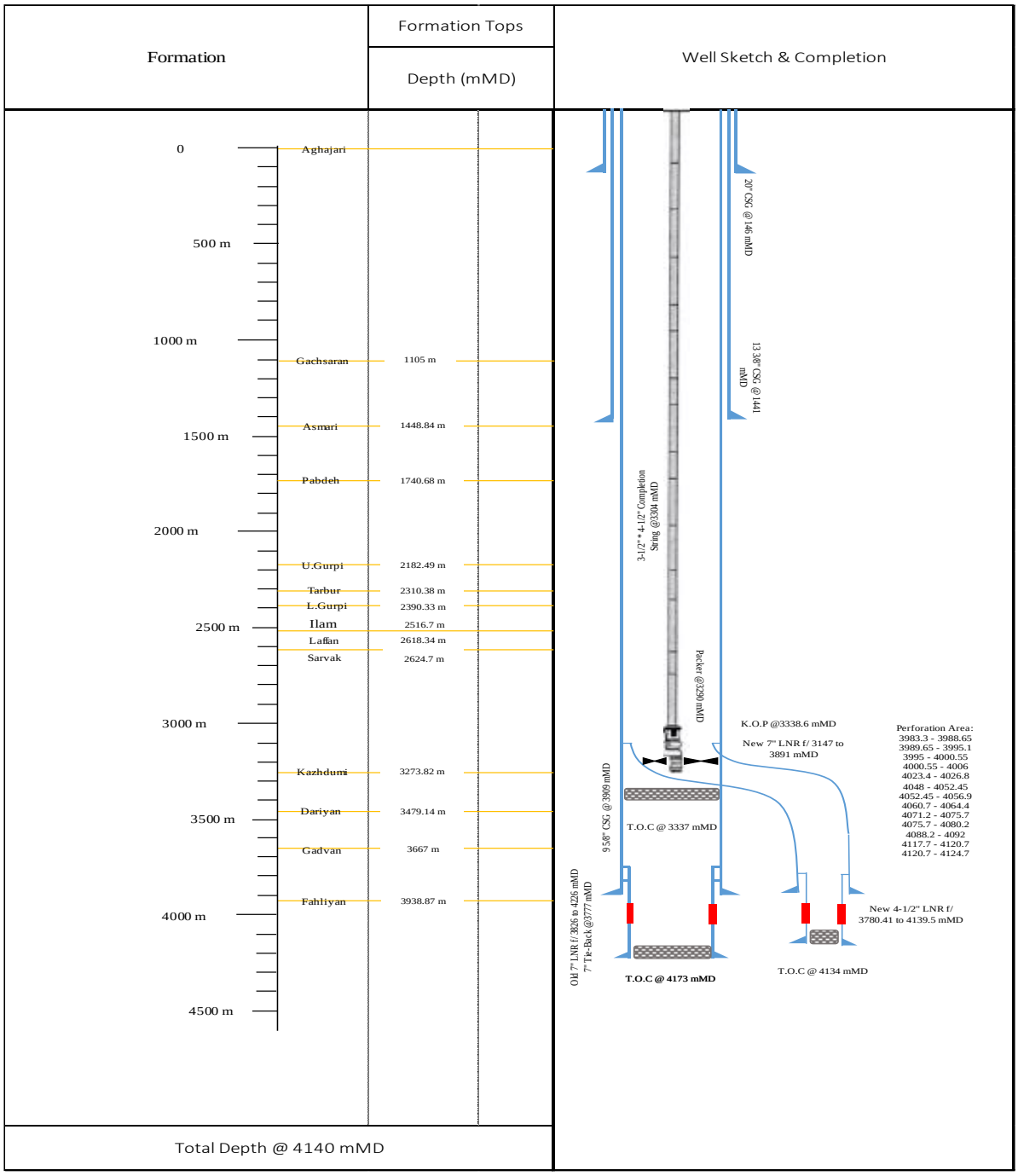


Figure 5: AZNS-127 Well Schematic After workover #2

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
		Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
Miocene							
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi / Tarbur / L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
		Laffan		Claystone	2691	2642	2625
Turonian Cenomanian		Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian		Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian		Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian		Fahliyan		Limestone, Marl	4092	3989	3974

Figure 6: Azadegan Stratigraphic Column

3. Logging Program

Hole Size	Casing Size	Formation	Logs
			Cased Hole
12 – 1/4"	9 – 5/8" CSG.	Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan	Corrosion Log
8 – 1/2"	7" Liner	Fahliyan	

Table 8: AZNS-127 Logging Program

4. Program Outline

4.1 Contractors Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 208
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

Table 9: AZNS-127 Contractor's Involve

4.2 Time Breakdown

Work-Over Operation TBD of Well: AZNS-127 Work-over #2 (Fahliyan Producer)	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over												
Held PJSM, Spud the Rig. Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.	De-Completion		Fa.	4140		12	0.50	0.5	0.6	13.2	0.55	0.6
R/U SL equipment, Remove 4 1/16" X.M.T cap, Install Lubricator & S/L BOP, RIH to 3SV depth, set protection sleeve, RIH with gauge ring to No-Go depth, POOH to surface. Note: If observe any obstruction during RIH with gauge ring Follow Contingency Plan#01	De-Completion		Fa.	4140		12	0.50	1.0	0.6	13.2	0.55	1.1
R/U slick line, RIH with shifting tools & Try to open SSD, POOH to surface & R/D Slick line. Note: if unable to open SSD Follow Contingency Plan#02	De-Completion		Fa.	4140		24	1.00	2.0	1.1	26.4	1.10	2.2
kill the well with Kill Mud 106 pcf, Observe the well, RIH with SL/ & retrieve protection sleeve.	De-Completion		Fa.	4140		24	1.00	3.0	1.1	26.4	1.10	3.3
Secure the well, Set TWCV inside tubing hanger & pressure test, N/D XMT & N/U BOP, open tie down screws, perform BOP pressure Test,	De-Completion		Fa.	4140		64	2.67	5.7	2.9	70.4	2.93	6.2
RIH with TBG hanger running tools and screw on TBG hanger. Note:if Tubing Hanger Thread damaged follow Contingency Plan#03	De-Completion		Fa.	4140		12	0.50	6.2	0.6	13.2	0.55	6.8
Unlatch Permanent PKR from anchor seal, Circulation to balance mud weight, pull out completion string precautionary due to probable trapped pressure. While POOH monitor the well and annulus to make sure the well is safe and over balance. (from 3308m). Note:if unable to unlatch completion string due to string pack off with huge amount of debries follow contingency plan#04 Note:if unable to unlatch completion string due to string wash out follow contingency plan#05 Note: if observe any damage on THS, follow contingency plan#6	De-Completion		Fa.	4140		142	5.92	12.1	6.5	156.2	6.51	13.3
P/U & M/U 8 ½" bit & RIH with clean out BHA (P/U DP's JT by JT) to 7" L.LAP (3147m), CBU, POOH to surface & L/D bit.	De-Completion		Fa.	4140		72	3.00	15.1	3.3	79.2	3.30	16.6
RIH with 6" M.M J Mill & BHA to top of permanent PKR (P/U required DC & DP Jt by Jt), Mill permanent packer, chase the packer to 4 1/2" L.Lap (or where ever PKR stopped), CBU, POOH to surface.	De-Completion		Fa.	4140		108	4.50	19.6	5.0	118.8	4.95	21.5
RIH with Taper Tap and catch the packer, recover completion packer.POOH To Surface. Note:if string stuck while Recovering Completion Packer follow contingency plan#07	De-Completion		Fa.	4140		70	2.92	22.5	3.2	77.0	3.21	24.8
P/U & M/U 5-7/8 Mill and clean out BHA (P/U required DP Jt by Jt) & RIH to 4 1/2" L.Lap (3780 m), CBU till shakers clean, POOH to surface. Note: Due to used Synthetic completion fluid in last work over operation & probable Corrosion in 9 5/8" CSG & 7" LNR, follow contingency plan#8 for corrosion log	De-Completion		Fa.	4140		60	2.50	25.0	2.8	66.0	2.75	27.5
M/U Tandem scraper BHA, RIH with tandem scrapers including 9-5/8" casing scraper & 7" liner scraper to 4 1/2" L.Lap (3780 m). Scrap 9-5/8" casing and 7" liner, POOH to surface and L/D same.	Trip		Fa.	4140		36	1.50	0.0	1.7	39.6	1.65	0.0
Change Ram & Test it, Install Shoting Nipple R/U W/L Logging, RIH & Perform Logging (Corrosion Logging) Inside 7" Liner, POOH to Surafce.	Log		Fa.	4140		24	1.00	0.0	1.1	26.4	1.10	0.0
RIH & Perform Logging (Corrosion Logging) Inside 9-5/8" Casing ,POOH to Surafce.	Log		Fa.	4140		24	1.00	0.0	1.1	26.4	1.10	0.0
P/U & M/U 3 1/2" M.M. J Mill & 2 7/8" DP's as Jt by Jt & RIH with clean out BHA to BTM (4134 m), C.B.U, Spot non solid completion fluid to 50m above 4 1/2" LNR lap, POOH to surface & L/D 2 7/8" DP as Jt by Jt.	Trip		Fa.	4140		108	4.50	29.5	5.0	118.8	4.95	32.5
RIH scraper and scrape 7" production packer setting area, displace the well with completion fluid, POOH to surface.	Trip		Fa.	4140		0	0.00	29.5	0.0	0.0	0.00	32.5
RIH w/ DP's and DC's, POOH & L/D same.	Trip		Fa.	4140		0	0.00	29.5	0.0	0.0	0.00	32.5
Run 10K Fahliyan completion string including CIV.	Completion		Fa.	4140		168	7.00	36.5	7.7	184.8	7.70	40.2
Set TWCV inside TBG HGR, N/D BOP's & install X.M.T.	Completion		Fa.	4140		58	2.42	38.9	2.7	63.8	2.66	42.8
Pressure test the X.M.T, Retrive TWCV.	Completion		Fa.	4140		18	0.75	39.7	0.8	19.8	0.83	43.6
Secure the well by set BPV in TBG Hanger, close all X.M.T valves except BMV, and Release the Rig.	Completion		Fa.	4140		8	0.33	40.0	0.4	8.8	0.37	44.0
Retrieve BPV, R/U Slick Line, Set protection sleeve, retrieve Prong & plug from NO GO landing nipple, POOH.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
RIH with S/L, install Plug & Prong into NO GO landing nipple, POOH. As per client request	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
RIH with S/L, retrieve protection sleeve.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		Fa.	4140		Rig Less	0.00	40.0	0.0	0.0	0.00	44.0
Total Duration of 6-1/8" Hole Section:							***40.0***	***40.0***			***44.0***	***44.0***
NOTE:												
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip			4140		20	0.83	40.8	0.9	22.0	0.92	44.9
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			4140		22	0.92	41.8	1.0	24.2	1.01	45.9
Total Duration of AZNS-127 Work-over #2 Well:								41.8				45.9

Contingency Plans TBD of Well: AZNS-127 Work-over #2	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed												
Contingency plan #1: if observe any obstruction while each run w/ Gauge ring, POOH to surface, R/D Slick Line, R/U CT, RIH with CT to top of perforated interval, displace the well from below packer with proper mud weight, POOH to surafce, R/D CT.	De-Completion		Fa.	4140		120	5.00	5.00	5.5	132.0	5.50	5.5
Contingency Plan#02 R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D, Displace well with heavy mud and try to kill the well,Circulate & Balance MW.	De-Completion		Fa.	4140		360	15.00	20.00	16.5	396.0	16.50	22.0
Contingency Plan#03 RIH w/Taper Tap & Try to tap in & engage w/Tubing Hanger	De-Completion		Fa.	4140		48	2.00	22.00	2.2	52.8	2.20	24.2
Contingency Plan#04 R/U & RIH with FPIT tools to check string free point,POOH to Surface & L/D.R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D.POOH Completion String from Coliding Depths in Several Runs.	Completion		Fa.	4140		288	12.00	34.00	13.2	316.8	13.20	37.4
Contingency Plan#05 Pull Completion String untill string will be part from weaked point then POOH & L/D String in Several attempt. It is assumed to part from 4- 1/2" expansion joint as weak point. Mill the remainder of fish & packer. Follow the rest of main plan.	Completion		Fa.	4140		608	25.33	59.33	27.9	668.8	27.87	65.3
contingency plan#06 : if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG).	Trip		Fa.	4140		240	10.00	69.33	11.0	264.0	11.00	76.3
contingency plan#07 RIH w/5-7/8" or 6" Mill & Mill the Packer & Tail in Several Runs, RIH with fishing tools, try to catch the fish, POOH to Surafce & L/D mill.	Trip		Fa.	4140		240	10.00	79.33	11.0	264.0	11.00	87.3
contingency plan#08 Perform scraper Trip, RIH & Perform Logging (Corrosion Logging) Inside 7" Liner & 9-5/8" Casing,POOH to Surafce.Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC. RIH scraper and scrape 7" production packer setting area, displace the well with completion fluid, POOH to surface & L/D all Tubulars	Trip		Fa.	4140		514	21.42	100.75	23.6	565.4	23.56	110.8
Total Duration of Contingency Plan AZNS-127 #2 Work-over Well:								100.8				110.8

5. Sequence of Operation

5.1 Main Objective

Decision has been made to de – complete he well & re – complete with new completion equipment.

5.2 Main plan

1. Held PJSM. Spud the Rig.
2. Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & bleed off any annulus pressure to zero upon client's approval and recheck the well status.
3. R/U Slick line equipment, remove 4 – 1/16" XMT Cap, install lubricator & slickline BOP, RIH to 3SV depth, set protection sleeve, RIH w/Gauge ring to NO – GO depth, POOH to surface.

   Note: If observe any obstruction during RIH with gauge ring, follow contingency plan #1.

4. R/U slickline, RIH with shifting tool and try to open SSD, POOH to surface and R/D slickline.

   Note: if unable to open SSD, follow contingency plan #2.

5. Kill the well with kill mud 106^{pcf}, observe the well. RIH with SL & retrieve protection sleeve.
6. Secure the well, set TWCV inside tubing hanger & pressure test.
7. N/D XMT & N/U BOP, open tie down screws, perform BOP pressure test.
8. RIH with TBG HGR running tool and screw on TBG HGR.

   Note: if TBG HGR thread damaged, follow contingency plan #3.

9. Unlatch permanent packer from anchor seal, perform circulation to balance Mw, pull out completion string precautionary due to probable trapped pressure. While POOH

monitor the well & annulus to make sure the well is safe & over balance. (from 3308[m])

Note: if unable to unlatch completion string due to string pack off with huge amount of debris, follow contingency plan #4.

Note: if unable to unlatch completion string due to string wash – out, follow contingency plan #5.

Note: if observe any damage on T.H.S, follow contingency plan #6.

10. P/U & M/U 8 – 1/2" bit & RIH with C/O BHA (P/U DPs JT by JT) to 7" liner lap @3147[m], CBU, POOH to surface & L/D bit.
11. RIH with 6" Metal Muncher Junk mill and BHA to top of permanent packer (P/U required DCs & DPs JT by JT), mill the packer, chase the packer to 4 – 1/2" liner lap (or wherever packer stopped), CBU, POOH to surface.
12. RIH with taper tap & catch the packer, recover completion packer, POOH to surface.

    Note: If string stuck while recovering completion packer, Follow contingency plan #7.

13. P/U & M/U 5 – 7/8" mill & C/O BHA (P/U required DPs JT by JT) & RIH to 4 – ½" liner lap @3780[m], CBU till shakers clean, POOH to surface.

    Note: Due to used Synthetic completion fluid in last workover operation & probable corrosion in 9 – 5/8" casing & 7" liner, follow contingency plan #8 for corrosion log.

14. M/U tandem scraper BHA, RIH with tandem scraper including 9 – 5/8" casing & 7" liner scraper to 4 – 1/2" liner lap @3780[m]. scrap 9 – 5/8" casing & 7" liner, POOH to surface & L/D same.
15. Change RAM & test it, install shooting nipple. R/U wireline logging tools, RIH & perform logging (corrosion log) inside 7" liner, POOH to surface.
16. RIH & perform logging (corrosion log) inside 9 – 5/8" casing, POOH to surface.
17. P/U & M/U 3 – 1/2" M.M.J. mill & 2 – 7/8" DPs JT by JT & RIH with C/O BHA to bottom @4134[m], CBU. Spot non – solid completion fluid to 50[m] above 4 – 1/2" liner lap, POOH to surface & L/D 2 – 7/8" DPs JT by JT.

18. RIH scraper & scrape 7" production packer setting depth, displace the well with completion fluid, POOH to surface.
19. RIH with DPs & DCs, POOH & L/D same.
20. Run 10k Fahliyan completion string including CIV.
21. Set TWCV inside TBG HGR, N/D BOP's & install XMT.
22. Pressure test the X.M.T, Retrieve TWCV.
23. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and **Release the Rig.**
24. Retrieve BPV, R/U Slick Line, set protection sleeve, retrieve prong & plug from NO GO landing nipple, POOH.
25. RIH with CT to bottom, displace the well with diesel or nitrogen & observe well, POOH.
26. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.
27. Flow the well as pre – acid clean up till acceptable BSW & reading stable flow rate on 2 different choke sizes for 72 hrs.
28. R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L.
29. RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve.
30. Flow the well as post – acid clean – up till acceptable BSW & reading stable flow rate on 2 different choke sizes for min 96 hrs.

    Note: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

    Note: Acidizing & post – acid clean – up maybe perform as rig – less job upon contractor request & client's approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.

31. RIH with S/L, set protection Sleeve, POOH & R/D SL.
32. RIH with C/T to bottom & displace the well with diesel, POOH.

33. RIH with S/L, install plug & prong into NO GO landing nipple, as per client's request. POOH.
34. RIH with S/L, retrieve protection sleeve.
35. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS – 127 Workover #2:

Based on operation situation in each step, the following contingency plans might be followed:

Contingency plan #1: if observe any obstruction while each run w/ Gauge Ring, POOH to surface, R/D slickline, R/U CT & RIH to top of perforated interval, displace the well from below packer with proper mud weight, POOH to surface. R/D CT.

Contingency plan #2: R/U & RIH with colliding tool & collide completion string from proper depth, POOH & L/D, displace the well with heavy mud and try to kill the well, circulate & balance Mw.

Contingency plan #3: RIH with taper tap & try to tap in & engage with TBG HGR.

Contingency plan #4: R/U & RIH with FPIT tool to check string free point, POOH to surface & L/D. R/U & RIH with colliding tool & collide completion string from proper depth, POOH & L/D. POOH completion string from colliding depth in several runs.

Contingency plan #5: pull completion string until string will be part from weak point, then POOH & L/D string in several attempts. It is assumed to part from 4 – 1/2" expansion joint as weak point. Mill the remained of fish & packer. Follow the rest of main plan.

Contingency plan #6: if observe any leakage & failure on T.H.S, secure the well by barriers (RTTS / CMT Plug).

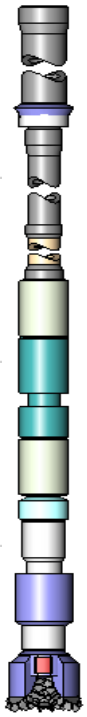
Contingency plan #7: RIH with 5 – 7/8" or 6" mill & mill the packer & tail in several runs, RIH with fishing tools, try to catch the fish, POOH to surface & L/D mill.

Contingency plan #8: perform scraper trip, RIH & perform logging (corrosion logging) inside 7" liner & 9 – 5/8" casing, POOH to surface, stand back required 5" & 3 – 1/2" DPs, 5" HWDPs & 6 – 1/2" DCs.
RIH scraper & scrape 7" production packer setting area, displace the well with completion fluid, POOH to surface & L/D all tubulars.

6. BHA Program

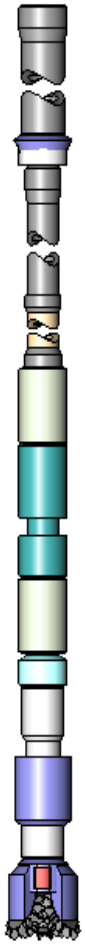
6.1 8-1/2" BHA

BIT SIZE (inch)	8-1/2"	BHA Configuration to C/Out Inside 9-5/8" Casing				Collar	OD
Depth In/Out (m)						DC-1	6-1/2"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	106						
B.FACTOR	0.784						
Weight in Air (lb)	22,041						
Bouyed Weight (lb)	17,273						
Bouyed Weight Below Jar (lb)	17,273						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	15	5	3	4-1/2" IF Pin x Box	137.1	138.15
	6-1/2" Bit Sub w/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x IF B	0.75	1.05
	8-1/2" MT Bit	1	8-1/2"	-	4-1/2" Reg. P	0.3	0.3

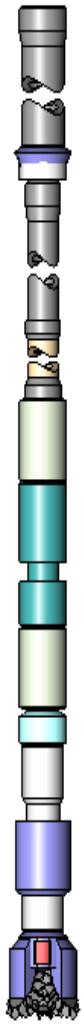
6.2 5-7/8" or 6" BHA

Mill SIZE (inch)	5-7/8" or 6"	**BHA Configuration to C/Out Inside 7" Liner**				**Collar**	**OD**
Depth In/Out (m)						DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	106						
B.FACTOR	0.784						
Weight in Air (lb)	25,440						
Bouyed Weight (lb)	19,936						
Bouyed Weight Below Jar (lb)	16,462						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	#REF!
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	#REF!
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	#REF!
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	6" Mill	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

6.1 3 – 1/2" BHA

Mill SIZE (inch)	3-1/2"	BHA Configuration to C/Out Inside 4-1/2" Liner	Collar	OD
Depth In/Out (m)			DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	103			
B.FACTOR	0.790			
Weight in Air (lb)	47,376			
Bouyed Weight (lb)	37,416			
Bouyed Weight Below Jar (lb)	19,909			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	1151.67
	3-1/2" Drill Pipe	74	3-1/2	2-7/16	3-1/2" IF P x B	710.4	1150.67
	XOS	1	3-1/2"	1-1/2"	3-1/2" IF B * 2-7/8" PAC PIN	0.5	440.27
	2-7/8" Drill Pipe	30	2-7/8"	1-1/2"	2-7/8" PAC P * B	274.2	439.77
	3-1/8" Drill Collar	18	3-1/8"	1-1/2"	2-7/8" PAC P * B	164.52	165.57
	3-1/8" Bit Sub	1	3-1/8"	1-1/2"	2-7/8" Reg. B x 2-7/8" PAC B	0.75	1.05
	3-1/2" Mill	1	3-1/2	1-1/2"	2-7/8" Reg. P	0.3	0.3

7. Mud Program

7.1 interval Description

De – Completion

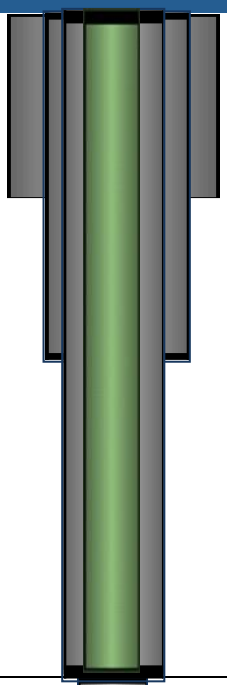
Hole Size (in)	Casing Size (in)	Casing Program	Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20”		146 m	**Already Has been Drilled**	
17 1/2”	13 3/8”		1441 m	**Already Has been Drilled**	
12 1/4”	9 5/8”		3909 m	**Already Has been Drilled**	
8 1/2”	7’’ Liner		From 3147 m To 3891 m	**Already Has been Drilled**	
6 1/8”	4-1/2” Liner		3780.41 to 4139.5	**Already Has been Drilled**	

Table 10: AZNS-127 Hole Schematic

7.1.1 Product Concentration & Function

103-106 pcf - Salt Saturated / Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic soda	0.5-1	Alkalinity Control
Soda Ash	1	Hardness Remover
Lime	As per Requirement	Alkalinity Control
Salt	109	non - solid weighting agent
Starch LV	4 - 6	Fluid loss controller
XC-Polymer	0.25-0.5	Viscosifier
Limestone	Up to 428	weighting agent
Biocide	As per Requirement	Anti-bacteria

MIXING PROCEDURE

1. Fill the tanks with calculated amount of drill water.
2. Add 109 ppb of NaCl to reach to saturation point at 75 pcf.
4. Raise the pH to 9 – 10 with addition of Caustic soda and Adjust calcium by Soda ash.
5. Add 5-6 ppb of Starch LV at rate of 5 min/Sack.
6. Add calculated amount of Limestone Powder to reach desire MW
7. Add calculated amount of Limestone Powder to reach desire MW
8. In case of foaming add required amount of Defoamer.
9. Recheck the pH and use Lime for pH adjustment.
10. Circulate the mud through gun line/Shearing unit at all the time to prevent any settling.
10- Biocide may be used as anti-bacteria agent to prevent polymer degradation.
11. Run a full test on mud and report the properties to office.

Table 11: AZNS-127 Product Concentration & Function

DRILLING FLUID SYSTEM	Salt Saturated / Polymer Mud
Key products	Salt (NaCl) , XC-Polymer, Caustic soda , Lime, Soda ash, Limestone
Water type	Drill Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

Table 12: AZNS-127 Drilling Fluid System

7.1.2 Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
Density	Pcf	103 - 106
Marsh Viscosity	Sec	45 - 55
PV	Cps	ALAP
Yield Point	lb/100ft²	14 – 20
API filter loss	cc/30min	<15
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	mg/l	< 1600
LGS	%	< 10
CL^-	Mg/l	18000-188,000

Table 13: AZNS-127 Drilling Fluid Properties

DRILLING FLUID TYPE			
Mud Density (pcf)	103 - 106	LGS Below (%vol)	0
Bit Size (inch)	0	Solid Control Equipment Eff. %	80
Section TD (m)	4226	Hole Wash Out %	0
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	0
Water Depth (m)		Meters To Drilled (m)	0

Volume information (Salt Pol Mud)		
Surface System	600	bbl
Cased Hole Volume	900	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,500	bbl
Dilution Volume	0	bbl
Sub Total	1,500	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	1,500	bbl
New Vol Required	1,500	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1	14
Caustic Soda	55	0.5 - 1	27
Salt (NaCl)	2,200	Up to 109	74
Starch LV	55	4 - 6	164
Limestone Powder	2,200	up to 428	292
Xanthan Gum	55	0.25 - 0.5	14

Contingency Materials			
Product Name	Function	Unit Size	QTY
Hematite	Pipe freeing agent	3300 lb	20
De foamer	De foamer	55 Gal	2
Shell fish F,M	Loss control material	55 lb	each size 60 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

Figure 7: AZNS-127 Mud Volume

7.2 interval Description

Completion

The completion fluid for this section will be Calcium Bromide with a scavenger, inhibitor.

7.2.1 Solid – Free $CaBr_2$ System

Calcium bromide A clear brine fluid, Calcium Bromide (CaBr2) solution is used as single salt brine in densities ranging from 11.7 lbs/gal to 15.1 lbs/gal. It is used as a completion, work over, gravel pack, and packer fluid. It can also be used as a spike fluid to increase density in fluids up to 14.2 lb/gal. Due to its inhibitive properties, CaBr2 can help reduce formation damage induced by clay swelling and dispersion.

Corrosion Control:

The corrosion rate of this type of fluid (CR) is less tha 5 MPY.

Pit management will be a key in the successful completion of this interval and sufficient time for planning and preparation should be allowed.

All valves should be checked for leakage beforehand and defects identified to be reported and rectified by drilling contractor.

The mud pits which are to be used to mix the completion brine, should emptied and cleaned adequately and associated lines flushed with water.

Filtration Process:

There is no need to focus on Filtration.

Displacement Process:

It is recommended to circulate and condition the mud prior to displacement and to spot the open hole section to conditioned mud or a solids-free fluid loss pill. Circulate and condition the mud as needed prior to pumping the clean out pills to optimum displacement properties).

After opening the circulating sub, the pumping sequence is as follows:

No.	Description	Volume Bbl	Comment
1	Displace to $CaBr_2$ at 103 to 106 pcf at highest rate available		
2	Continue pumping Completion Fluid until cleanliness specification has been met. Dump return at surface		

Completion Fluid Making Procedure & Consumption:

- Fill up fresh tank as require Calcium Bromide
- Cyr-Corrosion: Corrosion inhibitor Packer fluid design at rate of 1 % V/V.
- Oxygen scavenger (to eliminate dissolved oxygen as source of corrosion) at rate of 0.5 % V/V.
- H2S Scavenger at rate of 0.5 % V/V.

Rig Clean – Up Checklist:

A. Areas to clean:

- Surface pits
- Clean the suction, return, settling & reserve pits
- Clean the sand traps
- Clean the ditches & troughs
- Clean the shakers area including all solids equipment & lines
- Clean the mixing hoppers
- Clean the cement unit
- Clean any other pits or equipment that the completion fluid may contact
- Cycle all valves in mixing system

B. Items to flush out:

- All mud pumps & suction lines
- All centrifuges & lines
- Chemical & weighting mixing lines in contact with the completion fluid
- Lines from the pit room to the trip tank
- Lines from the rig floor to the trip tank
- Choke, choke manifold & kill line
- Standpipes
- Degassing & solid control equipment
- Lines to the cement unit and/or other pumping equipment
- Casing fill – up lines
- Lines to & from the filtration line
- Any other lines that the completion fluid may contact

Displacement Procedure:

1. Run bit and dual or tandem scrapers.2. Circulate and condition mud. Reduce funnel viscosity as much as possible while maintaining adequate rheological properties to clean hole and prevent settling of solids. Maintain critical annular velocity to assure that liner tops are thoroughly cleaned prior to displacement.

2. Rig up equipment for displacement of completion fluid. All tanks, circulating lines, and pumps to be used in the displacement should be cleared of residual mud.

3. Insure sufficient space is available to store displaced mud.

DRILLING FLUID TYPE					
Mud Density (pcf)	103	-	106	LGS Below (%vol)	
Bit Size (inch)		0		Solid Control Equipment Eff. %	
Section TD (m)		4226		Hole Wash Out %	
Casing OD/ID (inch)				Dilution Factor (bbl/m)	
Water Depth (m)				Meters To Drilled (m)	

Volume information (CaBr2)		
Surface System	200	bbl
Cased Hole Volume	900	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,100	bbl
Dilution Volume	0	bbl
Sub Total	1,100	bbl
10 % Extera Vol	110	bbl
Volume Received	0	bbl
Sub Total	1,210	bbl
New Vol Required	1,210	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
CaBr2	55	36.5 gpb	803
Cacl2 (94%-97%)	2,200	202-209	13
Corrosion Inhibitor	55 Gal	1% V/V	9
H_2S Scavenger	55 Gal	0.5% V/V	5
O_2 Scavenger	55 Gal	0.5% V/V	5

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

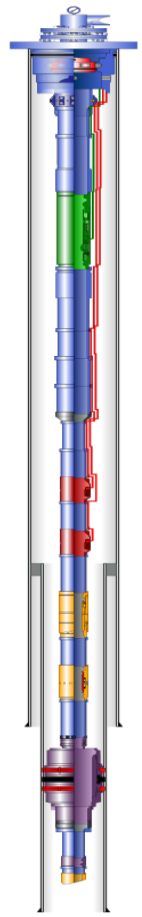
Figure 8: AZNS-127 Completion Fluid Volume

8. Completion

8.1 Proposed Completion string

AZNS-127 Workover #2 Completion Sketch

Contractor		Liner & Tubing Data					Client		
Field	AZADEGAN	Description	7" Liner	4 1/2" Liner	3 1/2" Tubing	4 1/2" Tubing	7" ST #01 Liner Lap		3147
Well Number:	AZNS-127	OD[in]	7.00	4.50	3.50	4.50	7" ST #01 Liner shoe		3891
Rig :	OEOC-201	Nominal weight [lb/ft]	29.00	13.50	10.20	18.90	4 1/2" Liner lap		3078.5
RTE-GLE	12.95	Grade	110 SS	110SS	L-80	L-80	4 1/2" Liner Shoe		4139.5
RTE-THF	3.95	Thread	TPCQ	TP-G2	HSM3	HSM3	Total Depth:		4140
Rig Spudded Date :	TBA	Inside diameter [in]	6.184	3.92	2.92	3.640	X-mass Tree:		MSP-T TYPE
End of Operation	TBA	API Drift Diameter	6.059	3.80	2.80	3.515	Size & Rating:		4 1/16"-10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)			11.788	11.788	0.000
	2	TBG HANGER 11" * 4 1/2'' ACME Box up * 4 1/2'' TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type ''H'' Two Way	5.118	3.64	0.52	12.308	11.788
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	6.1	3.64	0.964	13.272	12.308
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.5	60.772	13.272
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.624	62.396	60.772
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5.5	3.64	1.726	64.122	62.396
	7	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type,	7.5	3.437	1.783	65.905	64.122
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5.5	3.64	1.117	67.022	65.905
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.624	68.646	67.022
	10	315 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	2992.5	3061.146	68.646
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.119	3062.265	3061.146
	12	17 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	161.5	3223.765	3062.265
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.56	3225.325	3223.765
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3226.055	3225.325
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3227.875	3226.055
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3237.375	3227.875
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3239.195	3237.375
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3239.925	3239.195
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.8	3241.725	3239.925
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3251.225	3241.725
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3253.045	3251.225
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5.135	2.922	5.755	3258.800	3253.045
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3260.622	3258.800
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3270.122	3260.622
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3271.747	3270.122
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	3272.937	3271.747
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.7	3274.637	3272.937
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.27	3283.907	3274.637
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	3285.632	3283.907
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4.000	2.922	0.23	3285.862	3285.632
	31	7" 29 PPF CASING HYDRAULIC SET RERMANENT PRODUCTION PACKER TO BE RUN IN 7.00" 29 PPF CASING, SUPER 13CR-110KSI	5.910	2.922	4.44	3290.302	3285.862
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.822	3292.124	3290.302
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3301.624	3292.124
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3303.449	3301.624
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.750" TYPE XN (NO-GO ID: 2.635'')	4.5	2.75	0.37	3303.819	3303.449
	38	RE-ENTERYY MULE SHOE	4.25	2.922	0.37	3304.189	3303.819
		END OF COMPLETION STRING					3304.189

Tubing Retrievable Saftey Valve	65.905	64.122
Traveling Joint	3258.800	3253.045
CIV 1	3226.055	3225.325
CIV 2	3239.925	3239.195
SSD	3272.937	3271.747
7'' HYDRAULIC SET PERMANENT PRODUCTION PACKER	3290.302	3285.862
NO-GO Landing Nipple	3303.819	3303.449
Entry Guide	3304.189	3303.819

Figure 9: AZNS-127 Proposed Completion Schematic

9. BOP, Wellhead & X-Mass Tree

9.1 Existing Wellhead & X-Mass Tree

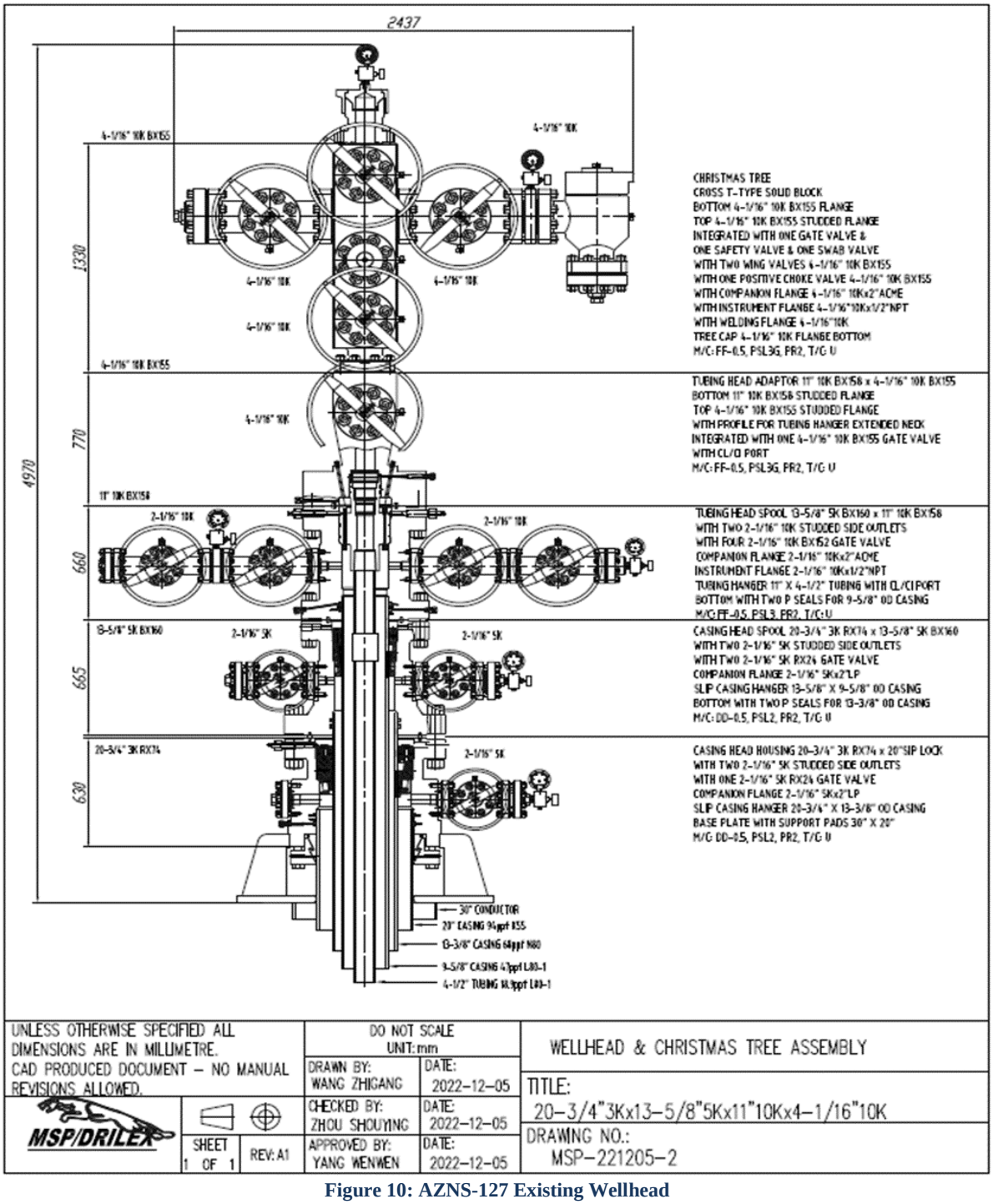


Figure 10: AZNS-127 Existing Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Sarvak	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Sarvak & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 14: AZNS-127 Wellhead Information

Wellhead Specification

NOTE: Wellhead and X-Mass Tree already installed by client.

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm** Material of body: Forged low alloy steel 410SS

	G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11” 10,000psi studded flange x 4 1/16” 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2” ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 15: AZNS-127 Wellhead Specification

9.2 BOP

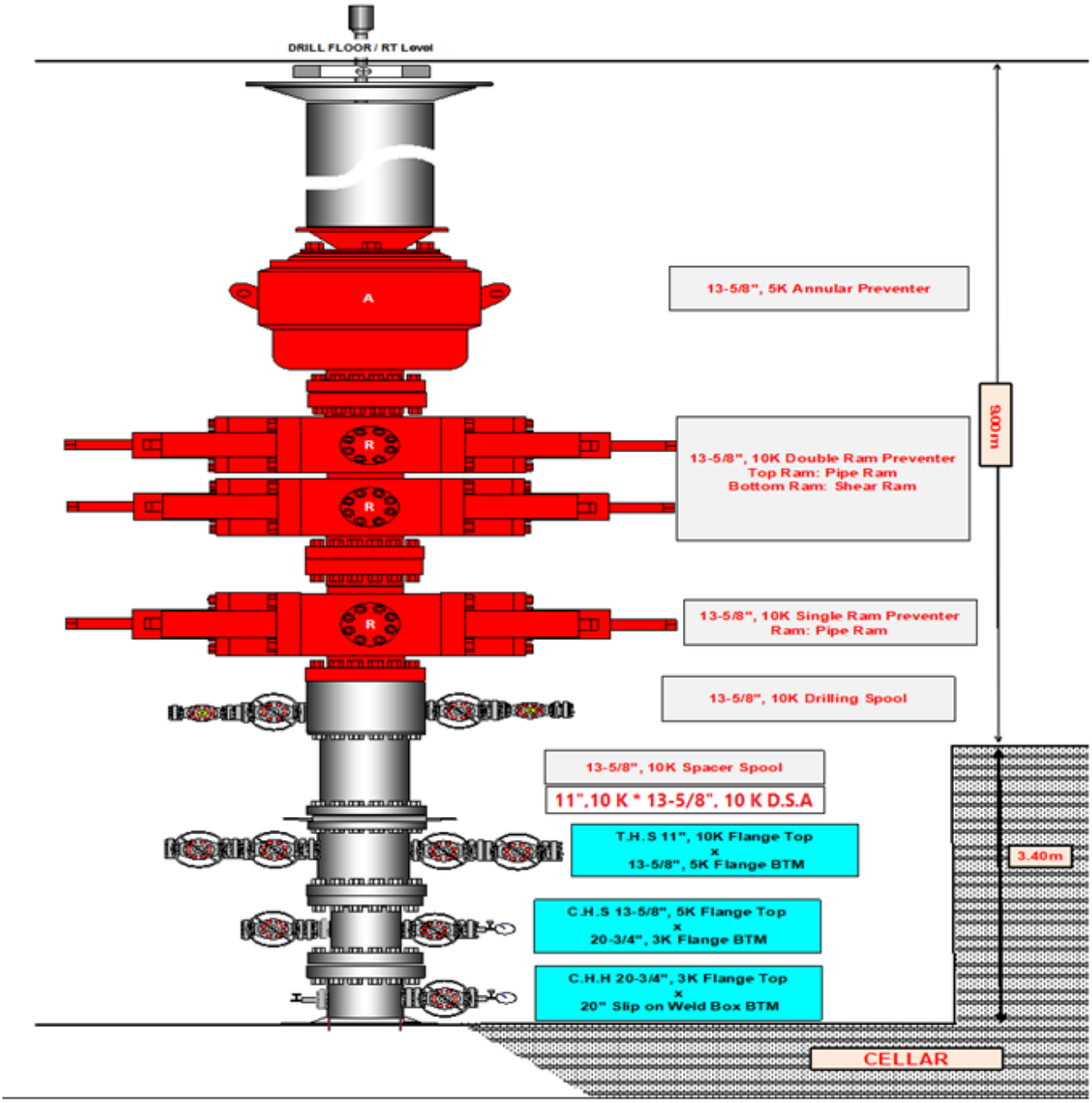


Figure 11: AZNS-127 BOP Sketch

10.

10. Waste Management

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

11. Health, Safety and Environments

11.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

11.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

11.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

11.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

11.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

11.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.

Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:

Once per tour.

Whenever changes occur in the mud density/rheology.

Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

11.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

11.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.

The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.

When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.

SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.

Once for every 12 hours in general;

After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.

Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:

Each time before POOH after drilling oil and gas formations.

Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.

POOH after killing well.

POOH after lowering mud weight in wellbore.

POOH after curing mud loss during drilling oil and gas formation.

POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).

Regarding the shut-in methods selection

Soft shut-in method is recommended for gas well;

Hard shut-in method is recommended for oil well;
Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;
Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.
Flow check shall be conducted regularly via trip tank as follows
Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)
POOH to last casing shoe
Before POOH BHA through BOP stack
Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.
During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.
When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.
During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.
During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.
Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud.
Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.
The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.
During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.
"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

11.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

11.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

11.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).

Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).

Capable enough to provide storage for the entire range of hazardous wastes generated during the work.

The area shall be provided with Spill Containment and means of preventing run-off to and from the area.

All waste containers to be stored shall be appropriately labelled.

Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

12. H_2S Monitoring & Prevention Measure

12.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

12.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

13. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

14. Requirements

Item	Description	QTY.	Unit	Supplier
1	8-1/2" Milled Tooth Bit	1	Ea.	MSA
2	5-7/8" or 6" Mill	5	Ea.	MSA
3	2-1/4"*4" Taper Tap	2	Ea.	MSA
4	4 1/2" * 3 1/2" Completion String 10K	1	set	MSA
5	9-5/8" & 7" Scraper	1+1	ea.	OEOC
6	5" DP,HWDP,8-1/2" DC,	as required		OEOC
7	Well Head Services	as required		Service Co.
8	Surface Well Test Services	as required		Service Co.
9	Completion Services	as required		Service Co.
10	Slick line Services	as required		Service Co.
11	Casing Running Services	as required		Service Co.
12	Wireline Logging	as required		Service Co.
13	Dry Test, DST, RTTS	as required		Service Co.
14	Cementing Services	as required		Service Co.
15	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
16	H2S Services	as required		Service Co.

17	Mud Logging Services	as required		Service Co.
18	Rig Services	as required		Service Co.
19	Drilling & Completion Fluid Services	as required		Service Co.
20				

Note: Quantity & Spec to be conformed with contract's SOW.
Note: In case of any shortage or discrepancy, the contract's SOW will be reference.

Table 16: AZNS-127 List of Requirement

15. Revision Corrections

Drilling Engineering							
۱	اسامی نفرات پروژه آزادگان شرکت پتروپارس ذکر گردد.	۲	CM	اصلاح گردید.			AP
۲	• RTE به ۱۲,۹۵ اصلاح گردد. • TD: به ۴۱۴۰ اصلاح گردد. • آخرین فاصله مشبک کاری: ۴۱۲۰,۷-۴۱۲۴,۷ اصلاح شود. • نوع چاه به side track اصلاح شود	۶۰-۷	CM	اصلاح شد.			AP
۳	Casing & Liner Depth • در جدول سایز حفره اضافه گردد. • عمق کفشک جداری ۲۰ " به ۱۴۶ متر اصلاح گردد. • گرید آستری ۷" به TPCQ اصلاح شود.	۳۶۰-۷	CM	اصلاح شد.			AP
٤	Well Schematic before Workover: • آخرین فاصله مشبک کاری: ۴۱۲۰,۷-۴۱۲۴,۷ اصلاح شود. • عمق کفشک جداری ۲۰ " به ۱۴۶ متر اصلاح گردد.	۳۸	CM	اصلاح گردید.			AP

5	Work-Over Operation TBD of Well: ردیف ۶، و آیتم ۳، قبل از قرارداد protection sleeve و ذکر گردد لوبریکتور نصب شود ، 4 1/16" XMT cap برداشته میشود.	۴۵-۴۳	CM	اصلاح گردید.			AP
6	به collide عملیات cont#2 ردیف ۴ در خصوص دلیل عدم یکپارچگی و بر قرار بودن ارتباط مورد تائید از متن SSD نبوده و میبایست در صورت عدم باز شدن زیر استفاده گردد: if unable to open SSD, displace the well with heavy mud weight, kill the well while monitoring return from section C (side valve)	۴۹-۴۳	CM	این چاه بر مبنای چاه CBS با توجه به اینکه وجود Cont. ۱۱۳ میباشد امکان تغییر متن ندارد و شایان ذکر است با وجود برقرار بودن ارتباط هم ممکن است امکان جابجایی سیال وجود نداشته باشد با توجه به اینکه دالیز چاه دارای فشار میباشد ، در صورت عدم توانایی جابجایی سیال با CT ادامه عملیات مطابق با برنامه ارائه شده صورت میپذیرد.	مورد تائید مشاور نمی COLLIDE عملیات باشد، و شایان ذکر است در صورت عدم باز و حتی عدم ، برای جابجایی SSD شدن توانایی جابجایی از بالا، می بایست با کویل تیوبینگ برای کشتن چاه با گل سنگین اقدام شود. با توجه به اینکه دالیز چاه دارای فشار می و عدم توانایی SSD باشد در صورت باز نشدن و جابجایی سیال از درون رشته با CT همچنین گردش معکوس، ادامه عملیات مطابق با برنامه ارائه شده بلامانع است.		AN
7	ردیف ۶ و ۷ در صفحه ۴۳ جدول و ۴۵ متن: قبل از نصب، protection sleeve ذکر گردد که TWCV خارج می گردد.	۴۵-۴۳	CM	اصلاح گردید. بعد از عملیات کشتن چاه و حصول اطمینان از پایان عملیات SL است در ردیف ۵ آورده شده	اصلاح نگردیده است. ذکر گردد: R/u SL equipment, remove protection sleeve, secure the well,….		AP

8	ردیف ۸: و ۱۰ در صفحه ۴۳ جدول و ۴۵ متن : ذکر گردد از انجام میشود.ذکر گردد که پکر anchor seal ، unlatch میباشد. همچنین متن به دلیل عدم وجود permanent فیش در چاه اصلاح شود.	۴۵-۴۳	CM	اصلاح گردید.			AP
9	ردیف ۱۲: یکبار در ردیف ۱۰ با میل تا لبه ۱/۲ ۴" رفته شده، TDدر این مرحله آیتم ۱۶ به جای آیتم ۱۲ جهت انجام قبل از انجام مراحل کروژن لاگ قرار گیرد. (در متن check sequence نیز اصلاح شود)	۴۶-۴۳	CM	به دلیل اینکه در حین انجام عملیات ردیف ۱۰ ”4 ½ قبل از رسیدن به لبه PKRممکن است متوقف شود لذا انجام ردیف ۱۶ ضروری میباشد.			AP
10	ذکر گردد که جابجایی سیال ”7 ردیف ۱۷: بعد از اسکرپ پکر نیز اصلاح sequence انجام میشود.(در متن TDتکمیلی از شود)	۴۷-۴۳	CM	هنگام gpmبه علت وجود محدودیت در جهت تمیز سازی ”7/8 2استفاده از لوله های contaminate و همچنین ایجاد ”4 ½لاینر پایین، لذا جابجایی سیال به gpmزیاد به دلیل صورت دو مرحله ای انجام میشود که مرتبه ”4 ½ تا لبه TDاول جابجایی سیال تکمیلی از ”4 ½ صورت خواهد گرفت و مرتبه دوم از لبه تا سطح انجام گردید.	انجام جابجایی سیال به صورت دو مرحلـه ای مانعی ندارد. ولی مـی بایسـت در برنامـه ذکـر گردد.		AP
11	درون TWCV و نصب تاج چاه، BOPردیف ۲۰: قبل از انداختن نیز اصلاح شود)sequence قرارگیرد..(در متن THS	۴۷-۴۳	CM	اصلاح گردید. اصلاح گردید.	بعد از نصب تـاج چـاه، قبـل از بیـرون آوردن تست فشار انجام گیرد. در متن بـر TWCV عکس ذکر شده.		AP
12	قبل از BPVردیف ۲۳: با توجه به ردیف ۲۲، و ایمن سازی چاه با نوشته TWCV ،BPVتریخص دکل، لذا در ردیف ۲۳ به جای نیز اصلاح شود)sequenceشود...(در متن	۴۷-۴۳	CM	اصلاح گردید اصلاح گردید.	در همین ردیف، ترتیب بیرون آوردن پرانگ و پلاگ اصلاح شود. ابتدا پرانگ نوشـته شـود و		AP

					سپس پلاگ.		
13	قبل از ایمن سازی چاه و NOGOردیف ۳۱: قرار دادن پلاگ درون (As per Client پایان کار ، باید بر اساس تصمیم کارفرما باشد . Approval(نیز اصلاح sequence در متن ذکر گردد.(در متن شود)	۴۷-۴۳	CM	همه چاه های قرارداد از جمله چاه CBSمطابق ۱۱۳ امکان اصلاح وجود ندارد. اصلاح گردید.	راندن کویل تیوبینگ چاه ۱۱۳CBSدر جهت جابجایی سیال قبل از کشتن چاه وجود نداشت، اما با توجه به شرایط قرار داده شد، با توجه به شرایط CBSلذا امکان تغییر در عملیاتی وجود دارد. اصلاح گردد.		AP
14	Cont #6: متن ذکر شده ناقص میباشد ، متن زیر جایگزین گردد: if observe any leakage and or failure on THS, after retrieving completion string secure the well by barriers (RTTS / CEMENT PLUG). Change THS, Pressure test the same, N/U BOP stack pressure test the same. • همچنین جمله تهیه ابزار از کارفرما حذف گردد	۴۹-۴۴	CM	شکست زمان بندی و هزینه های چاه مطابق ۱۱۳ که مبنای محاسبات چاه ۱۲۷ میباشد و با عنایت به عدم منظور گردیدن زمان و شرح عملیات تعویض THSدر متن الحاقی به چاه ۱۱۳) امکان اعمال متن CBSقرارداد (کارفرمای محترم با توجه به عواقب زمانی و مالی مقدور نمیباشد اصلاح گردید.	۱۱۰ روز زمان دیده شده است. توضیح Cont #6 پیمانکار توجیحی ندارد. علاوه بر این در خصوص متن ناقص و نا مفهوم، کامنت گذاشته شده است که می بایست اصلاح شود و مشخص گردد بعد از ایمن سازی چاه قرار است چه عملیاتی رخ دهد. بنابراین متن جهت شفاف سازی اصلاح گردد. همچنین جمله تهیه ابزار از کارفرما حذف گردد.		AP
15	Cont #8: سایز اسکرپر تریپ ذکر گردد ، همچنین جابجایی سیال انجام گردد.TDتکمیلی از همچنین جمله زیر حذف گردد. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.	۴۹-۴۴	CM	• به علت وجود محدودیت در gpm هنگام استفاده از لوله های 2 7/8” جهت تمیز سازی لاینر 4 ½” و همچنین ایجاد contaminate زیاد به دلیل gpm پایین، لذا جابجایی سیال	انجام جابجایی سیال به صورت دو مرحله ای مانعی ندارد. لذا میبایست در برنامه ذکر گردد.		AP

				به صورت دو مرحله ای انجام میشود که مرتبه اول جابجایی سیال تکمیلی از TD تا لبه 4 ½” صورت خواهد گرفت و مرتبه دوم از لبه 4 ½” تا سطح • امکان حذف جمله Stand . Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC به دلیل انتقال از متن اصلی به Cont مطابق نظر کارفرمای محترم فراهم نمیباشد و زمانی برای آن به صورت جداگانه منظور نگردیده • انجام گردید.			
۱6	تمیزسازی درون آستری ۱/۲ ۴" ارائه گرددBHA	۵۱	CM	اصلاح گردید.			AP
17	۷,۱interval Description • سایز جدای ۵/۸ ۱۸" به ۲۰ " اصلاح و عمق آن به ۱۴۶ متر تغییر یابد. • عمق آستری ۷" اصلاح شود. • آستری ۱/۲ ۴" اضافه گردد.	52	CM	اصلاح گردید.			AP

۱8	Proposed Completion string: • نوع متریال پکر ذکر گردد که 13CR می باشد. • NOGO ID در جدول ۲,۵۶۲ ذکر شده، اما در متن ۲,۴۴" خورده است.یکسان سازی شود	۶۰	CM	• اصلاح گردید. • مطابق با اسناد ارائه شده توسط شرکت سازنده، minimum NOGO ID معادل ۲,۶۳۵" و Seal Bore ID معادل ۲,۷۵۰" میباشد.			AP
۱9	, اصلاح PSL-3 , به PSL-3G از THSدر جدول مشخصات گردد	۶۲	CM	اصلاح گردید.			AP
Fluid Engineering							
۱	وزن معادلkill mud.ذکر گردد		CM	اضافه گردید.			AP
۲	در صورت نیاز به بالا بردن وزن و استفاده از فروبار ، گرفتن تائیدیه از کارفرما الزامیست.		CM	در صورت نیاز اخذ خواهد شد.			AN
۳	برنامه پسماند ارائه نگردیده است. میبایست در برنامه یا به صورت مجزا ارسال گردد.	۶۶	CM	به صورت جداگانه قبل از شروع عملیات ارائه می‌گردد.			AN